Exponential length lower bounds for binary two-query relaxed locally decodable codes

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Editorial scope. Counted principal result: thm:main-2qRLDC. The weak/strong relaxed definitions and Hamming/insdel error models are preserved separately. Other source-local results and strong-LCC application consequences are uncounted context.

Allowed standard prerequisites (editorial).

- Standard LDC lower bounds and cited concatenation/coding primitives. Exact imported statements and full original bibliography retained; all local reductions, encoder, decoder and synchronization proofs included.

1 Introduction

Locally Decodable Codes (LDCs) [KT00, STV99] are error-correcting codes $C : \Sigma^n \to \Sigma^m$ that have super-fast decoding algorithms that can recover individual symbols of a *message* $x \in \Sigma^n$, even when worst-case errors are introduced in the *codeword* $C(x)$. Similarly, Locally Correctable Codes (LCCs) are error-correcting codes $C : \Sigma^n \to \Sigma^m$ for which there exist very fast decoding algorithms that recover individual symbols of the *codeword* $C(x) \in \Sigma^m$, even when worst-case errors are introduced. LDCs/LCCs were first discovered by Katz and Trevisan [KT00] and since then have proven to be crucial tools in many areas of computer science, including private information retrieval, probabilistically checkable proofs, self-correction, fault-tolerant circuits, hardness amplification, and data structures (e.g., [BFLS91, LFKN92, BLR93, BK95, CKGS98, CGdW13, ALRW17] and surveys [Tre04, Gas04]).

The *parameters* of interest of these codes are their *rate*, defined as the ratio between the message length n and the codeword length m, their *relative minimum distance*, defined as the minimum normalized Hamming distance between any pair of codewords, and their *locality* or *query complexity*, defined as the number of queries a decoder makes to a received word $y \in \Sigma^m$. Trade-offs between the achievable parameters of Hamming LDCs/LCCs have been studied extensively over the last two decades [KdW04, WdW05, GKST06, Woo07, Yek08, Yek12, DGY11, Efr12, GM12, BDSS16, BG17, BGT17, DSW17, KMRS17, BCG20] (see also surveys by Yekhanin [Yek12] and by Kopparty and Saraf [KS16]).

Specifically, for 2-query Hamming LDCs/LCCs it is known that $m = 2^{\Theta(n)}$ [KdW04, GKST06, BRdW08, BGT17]. However, for $q > 2$ queries, the current gap between upper and lower bounds is superpolynomial in n. In particular, the best constructions have super-polynomial codeword length [Yek08, DGY11, Efr12], while the most general lower bounds for $q \geq 3$ are of the form $m = \Omega((\frac{n}{\log n})^{1+1/(\lceil \frac{q}{2} \rceil - 1)})$ [KT00, KdW04]. In particular, for $q = 3$, [KT00] showed an $m = \Omega(n^{3/2})$ bound, which was improved in [KdW04] to $m = \Omega(n^2/\log^2 n)$. This was further improved by [Woo07, Woo12] to $m = \Omega(n^2/\log n)$ for general codes and $m = \Omega(n^2)$ for linear codes. [BGT17] used new combinatorial techniques to obtain the same $m = \Omega(n^2/\log n)$ bound. A very recent paper [AGKM22] breaks the quadratic barrier and proves that $m = \Omega(n^3/\text{poly}\log n)$. We note that the exponential lower bound on the length of 3-query LDCs from [GM12] holds only for some restricted parameter regimes, and do not apply to the natural ranges of the known upper bounds.

Motivated by this large gap in the constant-query regime, as well as by applications in constructions of Probabilistically Checkable Proofs (PCPs), Ben-Sasson, Goldreich, Harsha, Sudan, and Vadhan [BGH+06] introduced a relaxed version of LDCs for Hamming errors. Specifically, the decoder is allowed to output a "decoding failure" answer (marked as "$\perp$"), as long as it errs with some small probability. More precisely, a $(q, \delta, \alpha, \rho)$-*relaxed LDC* is an error-correcting code satisfying the following properties.

Definition 1. *A $(q, \delta, \alpha, \rho)$-Relaxed Locally Decodable Code $C : \Sigma^n \to \Sigma^m$ is a code for which there exists a decoder that makes at most q queries to the received word y, and satisfies the following further properties:*

1. *(Perfect completeness) For every $i \in [n]$, if $y = C(x)$ for some message x then the decoder, on input i, outputs x_i with probability* 1.
2. *(Relaxed decoding) For every $i \in [n]$, if y is such that $dist(y, C(x)) \leq \delta$ for some unique $C(x)$, then the decoder, on input i, outputs x_i or $\perp$ with probability $\geq \alpha$.*
3. *(Success rate) For every y such that $dist(y, C(x)) \leq \delta$ for some unique $C(x)$, there is a set I of size $\geq \rho n$ such that for every $i \in I$ the decoder, on input i, correctly outputs x_i with probability $\geq \alpha$.*

We will call an RLDC that satisfies all 3 *conditions by the notion of* strong *RLDC, and one that satisfies just the first* 2 *conditions by the notion of* weak *RLDC, in which case it is called a (q, δ, α)-RLDC. Furthermore, if the q queries are made in advance, before seeing entries of the codeword, then the decoder is said to be* non-adaptive*; otherwise, it is called* adaptive*.*

The above definition is quite general, in the sense that $dist(a, b)$ can refer to several different distance metrics. In the most natural setting, we use $dist(a, b)$ to mean the "relative" Hamming distance between $a, b \in \Sigma^m$, namely $dist(a, b) = |\{i : a_i \neq b_i\}|/m$. This corresponds to the standard RLDCs for Hamming errors. As it will be clear from the context, we also use $dist(a, b)$ to mean the "relative" Edit distance between $a, b \in \Sigma^*$, namely $dist(a, b) = \mathsf{ED}(a, b)/(|a| + |b|)$, where $\mathsf{ED}(a, b)$ is the minimum number of insertions and deletions to transform string a into b. This corresponds to the new notion introduced and studied here, which we call *Insdel RLDCs*. Throughout this paper, we only consider the case where $\Sigma = \{0, 1\}$.

Definition 1 has also been extended recently to the notion of *Relaxed Locally Correctable Codes (RLCCs)* by Gur, Ramnarayan, and Rothblum [GRR20]. RLDCs and RLCCs have been studied in a sequence of exciting works, where new upper and lower bounds have emerged, and new applications to probabilistic proof systems have been discovered [GL19a, CGS20, GRR20, AS21, GL21].

Surprisingly, Ben-Sasson et al. [BGH+06] construct strong RLDCs with $q = O(1)$ queries and $m = n^{1+O(1/\sqrt{q})}$, and more recently Asadi and Shinkar [AS21] improve the bounds to $m = n^{1+O(1/q)}$, in stark contrast with the state-of-the-art constructions of standard LDCs. Gur and Lachish [GL21] show that these bounds are in fact tight, as for every $q \geq 2$, every weak q-query RLDC must have length $m = n^{1+1/O(q^2)}$ for non-adaptive decoders. We remark that the lower bounds of [GL21] hold even when the decoder does not have perfect completeness and in particular valid message bits are decoded with success probability 2/3. Dall'Agnon, Gur, and Lachish [DGL21] further extend these bounds to the setting where the decoder is adaptive, with $m = n^{1+1/O(q^2 \log^2 q)}$.

1.1 Our results

As discussed before, since the introduction of RLDCs, unlike standard LDCs, they displayed a behaviour amenable to nearly linear-size constructions, with almost matching upper and lower bounds. However, recently [GL21] conjecture that for $q = 2$ queries, there is in fact an exponential lower bound, matching the bounds for standard LDCs.

In this paper, our first contribution is a proof of their conjecture, namely to show that Hamming 2-query RLDCs require exponential length. In fact, our exponential lower bound for $q = 2$ applies even to weak RLDCs, which only satisfy the first two properties (perfect completeness and relaxed decoding), and even for adaptive decoders.

Theorem 1. *Let $C\colon \{0,1\}^n \to \{0,1\}^m$ be a weak adaptive $(2, \delta, 1/2 + \varepsilon)$-RLDC. Then $m = 2^{\Omega_{\delta,\varepsilon}(n)}$.*

Our results are the first exponential bounds for RLDCs. Furthermore, combined with the constructions with nearly linear codeword length for some constant number of queries [BGH+06, AS21], our results imply that RLDCs experience a "phase transition"-type phenomena, where the codeword length drops from being exponential at $q = 2$ queries to being almost linear at $q = c$ queries for some constant $c > 2$. In particular, this also implies that there is a query number q where the codeword length drops from being super-polynomial at q to being polynomial at $q + 1$. Finding this exact threshold query complexity is an intriguing open question.

As our second contribution, we introduce and study the notion of RLDCs correcting *insertions and deletions*, namely Insdel RLDCs. The non-relaxed variants of Insdel LDCs were first introduced in [OPC15], and were further studied in [BBG+20,CLZ20,BB21]. Local decoding in the Insdel setting is motivated in DNA storage [YGM17], and in particular [BSB+21] show recent advances in bio-technological aspects of random access to data in these precise settings.

In [OPC15,BBG+20], the authors give Hamming to Insdel reductions which transform any Hamming LDC into an Insdel LDC with rate reduced by a constant multiplicative factor, and locality increased by a polylog(m) multiplicative factor. Unfortunately, these compilers do not imply constant-query Insdel LDCs, whose existence is still an open question.

The results of [BCG+22] show strong lower bounds on the length of constant-query Insdel LDCs. In particular, they show that linear Insdel LDCs with 2 queries do no exist, general Insdel LDCs for $q = 3$ queries must have $m = \exp(\Omega(\sqrt{n}))$, and for $q \geq 4$ they must have $m = \exp(n^{\Omega(1/q)})$.

In this work we continue the study of locally decodable codes in insertion and deletion channels by proving the first upper and lower bounds regarding the relaxed variants of Insdel LDCs. We first consider strong Insdel RLDCs, which satisfy all three properties of Definition 1 and where the notion of distance is now that of relative edit distance. We adapt and extend the results of [BCG+22] to establish strong lower bounds on the codeword length of strong Insdel RLDCs. In particular, we prove that $m = \exp(n^{\Omega(1/q)})$ for any strong Insdel RLDC with locality q.

Theorem 2. *Let $C\colon \{0,1\}^n \to \{0,1\}^m$ be a non-adaptive strong $(q, \delta, 1/2 + \beta, \rho)$-Insdel RLDC where $\beta > 0$. Then for every $q \geq 2$ there is a constant $c_1 = c_1(q, \delta, \beta, \rho)$ such that*

$$m = \exp\left(c_1 \cdot n^{\Omega_\rho(\beta^2/q)}\right).$$

Furthermore, the same bound holds even if C does not have perfect completeness. If C has an adaptive decoder, the same bound holds with β replaced by $\beta/2^{q-1}$. Formally, there exists a constant $c_2 = c_1(q, \delta, \beta/2^{q-1}, \rho)$ such that

$$m = \exp\left(c_2 \cdot n^{\Omega_\rho(\beta^2/(q2^{2q}))}\right).$$

Our reduction shown in the proof of Theorem 1, together with the impossibility results of standard *linear* or *affine* 2-query Insdel LDCs from [BCG+22] show a further impossibility result for linear and for affine 2-query Insdel RLDCs (see remarks before Corollary 2). A linear code of length m is defined over a finite field $\mathbb{F}$ and it is a linear subspace of the vector space $\mathbb{F}^m$, while an affine code is an affine subspace of $\mathbb{F}^m$.

We then consider *weak* Insdel RLDCs that only satisfy the first two properties (perfect completeness and relaxed decoding). In contrast with Theorem 2, we construct weak Insdel RLDCs with constant locality $q = O(1)$ and length $m = n^{1+\gamma}$ for some constant $\gamma \in (0, 1)$. To the best of our knowledge, this is the first positive result in the constant-query regime and the Insdel setting. However, the existence of a constant-query standard Insdel LDC (or even a constant-query strong Insdel RLDC) with any rate remains an open question. Finally, it is easy to see that our exponential lower bound for weak Hamming RLDCs with locality $q = 2$ still applies in the Insdel setting, since Insdel errors are more general than Hamming error. Thus, in the Insdel setting we discover the same "phase transition"-type phenomena as for Hamming RLDCs.

Theorem 3. *For any $\gamma > 0$ and $\varepsilon \in (0, 1/2)$, there exist constants $\delta \in (0, 1/2)$ and $q = q(\delta, \varepsilon, \gamma)$, and non-adaptive weak $(q, \delta, 1/2 + \varepsilon)$-Insdel RLDCs $C\colon \{0,1\}^n \to \{0,1\}^m$ with $m = O(n^{1+\gamma})$.*

We remark that in the Hamming setting, [BGH+06] shows that the first two properties of Definition 1 imply the third property for codes with constant query complexity and which can withstand a constant

fraction of errors. Our results demonstrate that, in general, unlike in the Hamming case, the first two properties do not imply the third property for Insdel RLDCs from Definition 1. Indeed, while for strong Insdel RLDCs we have $m = \exp(n^{\Omega(1/q)})$ for codes of locality q, there exists $q = O(1)$ for which we have constructions of weak Insdel RLDCs with $m = n^{1+\gamma}$. This observation suggests that there are significant differences between Hamming RLDCs and Insdel RLDCs.

We note that our construction of weak Insdel RLDCs can be modified to obtain strong Insdel Relaxed Locally Correctable Codes (Insdel RLCCs). Informally, an Insdel RLCC is a code for which codeword entries can be decoded to the correct value or $\perp$ with high probability, even in the presence of insdel errors. The formal definition of RLCC is given in Section 6.3 (see Definition 4). We have the following corollary.

Corollary 1. *For any $\gamma > 0$ and $\varepsilon \in (0, 1/2)$, there exist constants $\delta \in (0, 1/2)$ and $q = q(\delta, \varepsilon, \gamma)$, and non-adaptive strong $(q, \delta, 1/2 + \varepsilon, 1/2)$-Insdel RLCCs $C\colon \{0,1\}^n \to \{0,1\}^m$ with $m = O(n^{1+\gamma})$.*

1.2 Overview of techniques

1.2.1 Exponential Lower Bound for Weak Hamming RLDCs with $q = 2$

To simplify the presentation, we assume a non-adaptive decoder in this overview. While the exact same arguments do not directly apply to adaptive decoders[1], with a bit more care they can be adapted to work in those settings.

At a high level we prove our lower bound by transforming any non-adaptive 2-query weak Hamming RLDC for messages of length n and δ fraction of errors into a standard 2-query Hamming LDC for messages of length $n' = \Omega(n)$, with slightly reduced error tolerance of $\delta/2$. Kerenidis and de Wolf [KdW04] proved that any 2-query Hamming LDC for messages of length n must have codeword length $m = \exp(\Omega(n))$. Combining this result with our transformation, it immediately follows that any 2-query weak Hamming RLDC must also have codeword length $m = \exp(\Omega(n))$. While our transformation does not need the third property (success rate) of a strong RLDC, we crucially rely on the property of *perfect completeness*, and that the decoder only makes $q = 2$ queries.

Let $C\colon \{0,1\}^n \to \{0,1\}^m$ be a weak $(2, \delta, 1/2 + \varepsilon)$-RLDC. For simplicity (and without loss of generality), let us assume the decoder Dec works as follows. For message x and input $i \in [n]$, the decoder non-adaptively makes 2 random queries $j, k \in [m]$, and outputs $f^i_{j,k}(y_j, y_k) \in \{0, 1, \perp\}$, where y_j, y_k are answers to the queries from a received word y, and $f^i_{j,k}\colon \{0,1\}^2 \to \{0,1,\perp\}$ is a deterministic function. When there is no error, we have $y_j = C(x)_j$ and $y_k = C(x)_k$.

We present the main ideas below, and refer the readers to Section 4 for full details.

Fixable codeword bits. The starting point of our proof is to take a closer look at those functions $f^i_{j,k}$ with $\perp$ entries in their truth tables. It turns out that when $f^i_{j,k}$ has at least one $\perp$ entry in the truth table, $C(x)_j$ can be fixed to a constant by setting either $x_i = 0$ or $x_i = 1$, and same for $C(x)_k$. To see this, note that the property of perfect completeness forces $f^i_{j,k}$ to be 0 or 1 whenever $x_i = 0$ or $x_i = 1$ and there is no error. Thus if neither $x_i = 0$ nor $x_i = 1$ fixes $C(x)_j$, then there must be two entries of 0 and two entries of 1 in the truth table of $f^i_{j,k}$, which leaves no space for $\perp$ (see Claim 1). Thus, when there is at least one $\perp$ entry in the truth table of $f^i_{j,k}$, we say that $C(x)_j$ and $C(x)_k$ are *fixable* by x_i.

This motivates the definition of the set S_i, which contains all indices $j \in [m]$ such that the codeword bits $C(x)_j$ are fixable by x_i; and the definition of T_j, the set of all indices $i \in [n]$ such that $C(x)_j$ is fixable by the message bits x_i. It is also natural to pay special attention to queries j, k that are not both contained in S_i, since in this case the function $f^i_{j,k}$ never outputs $\perp$.

The query structure. In general, a query set $\{j, k\}$ falls into one of the following three cases: (1) both j, k lie inside S_i; (2) both j, k lie outside of S_i; (3) one of them lies inside S_i and the other lies outside

[1] For standard LDCs Katz and Trevisan [KT00] observed that an adaptive decoder could be converted into a non-adaptive decoder by randomly guessing the output y_j of the first query j to learn the second query k. Now we non-adaptively query the received codeword for both y_j and y_k. If our guess for y_j was correct then we continue simulating the adaptive decoder. Otherwise, we simply guess the output x_i. If the adaptive decoder succeeds with probability at least $p \geq 1/2 + \epsilon$ then the non-adaptive decoder succeeds with probability $p' \geq 1/4 + p/2 \geq 1/2 + \epsilon/2$. Unfortunately, this reduction does not preserve perfect completeness as required by our proofs for relaxed 2-query Hamming RLDCs i.e., if $p = 1$ then $p' = 3/4$.

of S_i. It turns out that case (3) essentially never occurs for a decoder with perfect completeness. The reason is that when, say, $j \in S_i$ and $k \notin S_i$, one can effectively pin down every entry in the truth table of $f^i_{j,k}$ by using the perfect completeness property, and observe that the output of $f^i_{j,k}$ does not depend on y_k at all (see Claim 2). Thus in this case we can equivalently view the decoder as only querying y_j where $j \in S_i$, which leads us back to case (1). In what follows, we denote by E_1 the event that case (1) occurs, and by E_2 the event that case (2) occurs.

The transformation by polarizing conditional success probabilities. We now give a high level description of our transformation from a weak RLDC to a standard LDC. Let y be a string which contains at most $\delta m/2$ errors from the codeword $C(x)$. We have established that the success probability of the weak RLDC decoder on y is an average of two conditional probabilities

$$\Pr[\mathsf{Dec}(i,y) \in \{x_i, \perp\}] = p_1 \cdot \Pr[\mathsf{Dec}(i,y) \in \{x_i,\perp\} \mid E_1] + p_2 \cdot \Pr[\mathsf{Dec}(i,y) \in \{x_i,\perp\} \mid E_2],$$

where $p_1 = \Pr[E_1]$ and $p_2 = \Pr[E_2]$. Let us assume for the moment that S_i has a small size, e.g., $|S_i| \le \delta m/2$. The idea in this step is to introduce additional errors to the S_i-portion of y, in a way that drops the conditional success probability $\Pr[\mathsf{Dec}(i,y) \in \{x_i,\perp\} \mid E_1]$ to 0 (see Lemma 5). In particular, we modify the bits in S_i to make it consistent with the encoding of any message $\hat{x}$ with $\hat{x}_i = 1 - x_i$. Perfect completeness thus forces the decoder to output $1 - x_i$ conditioned on E_1. Note that we have introduced at most $\delta m/2 + |S_i| \le \delta m$ errors in total, meaning that the decoder should still have an overall success probability of $1/2 + \varepsilon$. Furthermore, now the conditional probability $\Pr[\mathsf{Dec}(i,y) \in \{x_i,\perp\} \mid E_2]$ takes all credits for the overall success probability. Combined with the observation that Dec never outputs $\perp$ given E_2, this suggests the following natural way to decode x_i in the sense of a standard LDC: sample queries j,k according to the conditional probability given E_2 (i.e., both j,k lie outside S_i) and output $f^i_{j,k}(y_j, y_k)$. This gives a decoding algorithm for standard LDC, with success probability $1/2 + \varepsilon$ and error tolerance $\delta m/2$ (see Lemma 6), modulo the assumption that $|S_i| \le \delta m/2$.

Upper bounding $|S_i|$. The final piece in our transformation from weak RLDC to standard LDC is to address the assumption that $|S_i| \le \delta m/2$. This turns out to be not true in general, but it would still suffice to prove that $|S_i| \le \delta m/2$ for $n' = \Omega(n)$ of the message bits i. If we could show that $|T_j|$ is small for most $j \in [m]$, then a double counting argument shows that $|S_i|$ is small for most $i \in [n]$. Unfortunately, if we had $C(x)_j = \bigwedge_{i=1}^n x_i$ for $m/2$ of the codeword bits j then we also have $|T_j| = n$ for $m/2$ codeword bits and $|S_i| \ge m/2 \ge \delta m/2$ for all message bits $i \in [n]$. We address this challenge by first arguing that any weak RLDC for n-bit messages can be transformed into another weak RLDC for $\Omega(n)$-bit messages for which we have $|T_j| \le 3\ln(8/\delta)$ for all but $\delta m/4$ codeword bits. The transformation works by fixing some of the message bits and then eliminating codeword bits that are fixed to constants. Intuitively, if some $C(x)_j$ is fixable by many message bits, it will have very low entropy (e.g., $C(x)_j$ is the AND of many message bits) and hence contain very little information and can (likely) be eliminated. We make this intuition rigorous through the idea of random restriction: for each $i \in [n]$, we fix $x_i = 0$, $x_i = 1$, or leave x_i free, each with probability $1/3$. The probability that $C(x)_j$ is not fixed to a constant is at most $(1 - 1/3)^{|T_j|} \le \delta/8$, provided that $|T_j| \ge 3\ln(8/\delta)$. After eliminating codeword bits that are fixed to constants, we show that with probability at least $1/2$ at most $\delta m/4$ codeword bits $C(x)_j$ with $|T_j| \ge 3\ln(8/\delta)$ survived[2]. Note that with high probability the random restriction leaves at least $n/6$ message bits free. Thus, there must exist a restriction which leaves at least $n/6$ message bits free ensuring that $|T_j| \ge 3\ln(8/\delta)$ for at most $\delta m/4$ of the remaining codeword bits $C(x)_j$. We can now apply the double counting argument to conclude that $|S_i| \le \delta m/2$ for $\Omega(n)$ message bits, completing the transformation.

Adaptive decoders For possibly adaptive decoders, we are going to follow the same proof strategy. The new idea and main difference is that we focus on the first query made by the decoder, which is always non-adaptive. We manage to show that the first query determines a similar query structure, which is the key to the transformation to a standard LDC. More details can be found in Section 4.2.

[2]We are oversimplifying a bit for ease of presentation. In particular, the random restriction process may cause a codeword bit $C(x)_j$ to be fixable by a new message bit x_i that did not belong to T_j before the restriction – We thank an anonymous reviewer for pointing this out to us. Nevertheless, for our purpose it is sufficient to eliminate codeword bits that initially have a large $|T_j|$. See the formal proof for more details.

1.2.2 Lower Bounds for Strong Insdel RLDCs

We recall that a strong Insdel RLDC C is a weak Insdel RLDC which satisfies an additional property: for every $x \in \{0,1\}^n$ and $y \in \{0,1\}^{m'}$ such that $\mathsf{ED}(C(x), y) \leq \delta \cdot 2m$, there exists a set $I_y \subseteq [n]$ of size $|I_y| \geq \rho n$ such that for every $i \in I_y$, we have $\Pr[\mathsf{Dec}(i, y) = x_i] \geq \alpha$. In other words, for ρ-fraction of the message bits, the decoder can correctly recover them with high probability, just like in a standard Insdel LDC. Towards obtaining a lower bound on the codeword length m, a natural idea would be to view C as a standard Insdel LDC just for that ρ-fraction of message bits, and then apply the exponential lower bound for standard Insdel LDCs from [BCG+22]. This idea would succeed if the message bits correctly decoded with high probability were the same for all potential corrupted codewords y. However, it could be the case that $i \in I_y$ for some strings y, whereas $i \notin I_{y'}$ for other strings y'. Indeed, allowing the set I_y to depend on y is the main reason why very short constant-query Hamming RLDCs exist.

We further develop this observation to obtain our lower bound. We use an averaging argument to show the existence of a *corruption-independent* set I of message bits with $|I| = \Omega(n)$, which the decoder can recover with high probability. To this end, we need to open the "black box" of the lower bound result of Blocki et al. [BCG+22]. The proof in [BCG+22] starts by constructing an error distribution $\mathcal{E}$ with several nice properties, and deduce the exponential lower bound based solely on the fact that the Insdel LDC should, on average (i.e., for a uniformly random message x), correctly recover each bit with high probability under $\mathcal{E}$ (see Theorem 5). One of the nice properties of $\mathcal{E}$ is that it is oblivious to the decoding algorithm Dec. Therefore, it makes sense to consider the average success rate against $\mathcal{E}$, i.e., $\Pr[\mathsf{Dec}(i, y) = x_i]$, where $i \in [n]$ is a uniformly random index, $x \in \{0,1\}^n$ is a uniformly random string, and y is a random string obtained by applying $\mathcal{E}$ to $C(x)$. By replacing $\perp$ with a uniformly random bit in the output of Dec, the average success rate is at least $\rho\alpha + (1-\rho)\alpha/2 = (1+\rho)\alpha/2$, since there is a ρ-fraction of indices for which Dec can correctly recover with probability α, and for the remaining $(1-\rho)$-fraction of indices the random guess provides an additional success rate of at least $\alpha/2$. Assuming α is sufficiently close to 1, which we can achieve by repeating the queries independently for a constant number of times and doing something similar to a majority vote, the average success rate against $\mathcal{E}$ is strictly above $1/2$. Therefore, there exist a constant fraction of indices for which the success rate against $\mathcal{E}$ is still strictly above $1/2$, and the number of queries remains a constant. This is sufficient for the purpose of applying the argument in [BCG+22] to get an exponential lower bound. Full details appear in Section 5.

1.2.3 Constant-Query Weak Insdel RLDC

Our construction of a constant query weak Insdel RLDC uses code concatenation and two building blocks: a weak Hamming RLDC (as the outer code) with constant query complexity, constant error-tolerance, and codeword length $k = O(n^{1+\gamma})$ for any $\gamma > 0$ [BGH+06], and the Schulman-Zuckerman [SZ99] (from now on denoted by SZ) Insdel codes[3] (as the inner code). We let $C_{\text{out}} \colon \{0,1\}^n \to \{0,1\}^k$ and $C_{\text{in}} \colon [k] \times \{0,1\} \to \{0,1\}^t$ denote the outer and inner codes, respectively. Our final concatenation code C will have codewords in $\{0,1\}^m$ for some m (to be determined shortly), will have constant query complexity, and will tolerate a constant fraction of Insdel errors.

Code construction. Given a message $x \in \{0,1\}^n$, we first apply the outer code to obtain a Hamming codeword $y = y_1 \circ \cdots \circ y_k = C_{\text{out}}(x)$ of length k, where each $y_i \in \{0,1\}$ denotes a single bit of the codeword. Then for each index i, we compute $c_i = C_{\text{in}}(i, y_i) \in \{0,1\}^t$ as the encoding of the message (i, y_i) via the inner code. Finally, we output the codeword $C(x) \coloneqq c_1 \circ 0^t \circ c_2 \circ \cdots \circ 0^t \circ c_k$, where 0^t denotes a string of t zeros (which we later refer to as a buffer). Note that the inner code is a constant-rate code, i.e., $t = O(\log(k))$, and has constant error-tolerance $\delta_{\text{in}} \in (0, 1/2)$. Thus, the final codeword has length $m \coloneqq (2t-1)k = O(k \log(k))$ bits. For any constant $\gamma > 0$ we have a constant query outer code with length $k = O(n^{1+\gamma})$. Plugging this into our construction we have codeword length $m = O(n^{1+\gamma} \log n)$ which is $O(n^{1+\gamma'})$ for any constant $\gamma' > \gamma$.

Decoding algorithm: intuition and challenges. Intuitively, our relaxed decoder will simulate the outer decoder. When the outer decoder requests y_i, the natural approach would be to find and decode

[3] In particular, these are classical/non-local codes.

the block c_i to obtain (i, y_i). There are two challenges in this approach. First, if there were insertions or deletions, then we do not know where the block c_i is located; moreover, searching for this block can potentially blow-up the query complexity by a multiplicative polylog(m) factor [OPC15, BBG+20]. Second, even if we knew where c_i were located, because $t = O(\log k)$ and we want the decoder to have constant locality, we cannot afford to recover the entire block c_i.

We address the first challenge by attempting to locate block c_i under the optimistic assumption that there are no corruptions. If we detect any corruptions, then we may immediately abort and output $\perp$ since our goal is only to obtain a weak Insdel RLDC. Assuming that there were no corruptions, we know exactly where the block c_i is located, and we know that c_i can only take on two possible values: it is either the inner encoding of $(i, 0)$ or the inner encoding of $(i, 1)$. If we find anything inconsistent with the inner encoding of either $(i, 0)$ or $(i, 1)$, then we can immediately output $\perp$.

Checking consistency with the inner encodings of $(i, 0)$ and $(i, 1)$ is exactly how we address the second challenge. In place of reading the entire block c_i, we instead only need to determine whether (1) c_i is (close to) the inner encoding of $(i, 0)$, (2) c_i is (close to) the inner encoding of $(i, 1)$, or (3) c_i is not close to either string. In either case (1) or case (2), we simply output the appropriate bit, and in case (3), we simply output $\perp$. Thus, our Insdel RLDC decoder simulates the outer decoder. Whenever the outer decoder request y_i, we determine the expected location for c_i, randomly sub-sample a constant number of indices from this block and compare with the inner encodings of $(i, 0)$ and $(i, 1)$ at the corresponding indices. To ensure perfect completeness, we always ensure that *at least one* of the sub-sampled indices is for a bit where the inner encodings of $(i, 0)$ and $(i, 1)$ differ. If there are no corruptions, then whenever the simulated outer decoder requests y_i we will always respond with the correct bit. Perfect completeness of our Insdel RLDC now follows immediately from the perfect completeness of the outer decoder. Choosing a constant number of indices to sub-sample ensures that the locality of our weak Insdel RLDC decoder is a constant multiplicative factor larger than the outer decoder, which gives our Insdel RLDC decoder constant locality overall.

Analysis of the decoding algorithm. The main technical challenge is proving that our Insdel RLDC still satisfies the second condition of Definition 1, when the received word is not a correct encoding of the message x. Recall that $c_i = C_{\text{in}}(i, y_i)$, and suppose $\widetilde{c}_i \neq c_i$ is the block of the received word that we are going to check for consistency with the inner encodings of $(i, 0)$ and $(i, 1)$. Then, the analysis of our decoder falls into three cases. In the first case, if $\widetilde{c}_i$ is not too corrupted (i.e., $\mathsf{ED}(\widetilde{c}_i, c_i)$ is not too large), then we can argue that the decoder outputs the correct bit y_i or $\perp$ with good probability. In the second case, if $\widetilde{c}_i$ has high edit distance from both $C_{\text{in}}(i, 0)$ and $C_{\text{in}}(i, 1)$, then we can argue that the decoder outputs $\perp$ with good probability. The third case is the most difficult case, which we describe as "dangerous". We say that the block $\widetilde{c}_i$ is *dangerous* if the edit distance between $\widetilde{c}_i$ and $C_{\text{in}}(i, 1 - y_i)$ is not too large; i.e., $\widetilde{c}_i$ is close to the encoding of the opposite bit $1 - y_i$.

The key insight to our decoding algorithm is that as long as the number of dangerous blocks $\widetilde{c}_i$ is upper bounded, then we can argue the overall probability that our decoder outputs y_i or $\perp$ satisfies the relaxed decoding condition of Definition 1. Intuitively, we can we think of our weak Insdel RLDC decoder as running the outer decoder on a string $\tilde{y} = \tilde{y}_1 \circ \ldots \circ \tilde{y}_k$, where each $\tilde{y}_i \in \{0, 1, \perp\}$ and the outer decoder has been modified to output $\perp$ whenever it queries for y_i and receives $\perp$. Observe that if δ_{out} is the error-tolerance of the outer decoder, then as long as the set $\left|\{i \ : \ \widetilde{y}_i \neq \perp \wedge \widetilde{y}_i \neq y_i\}\right| \leq \delta_{\text{out}} k$, the modified outer decoder, on input $j \in [n]$, will output either the correct value x_j or $\perp$ with high probability (for appropriate choices of parameters). Intuitively, if a block is "dangerous" then we can view $\widetilde{y}_i = 1 - y_i$, and otherwise we have $\widetilde{y}_i \in \{y_i, \perp\}$ with reasonably high probability. Thus, as long as the number of "dangerous" block is at most $\delta_{\text{out}} k/2$, then our relaxed Insdel decoder will satisfy the second property of Definition 1 and output either x_j or $\perp$ with high probability for any $j \in [n]$.

Upper bounding the number of dangerous blocks. To upper bound the number of "dangerous" blocks we utilize a matching argument based on the longest common sub-sequence (LCS) between the original codeword and the received (corrupted) word. Our matching argument utilizes a key feature of the SZ Insdel code. In particular, the Hamming weight (i.e., number of non-zero symbols) of every substring c' of an SZ codeword is at least $\lfloor |c'|/2 \rfloor$. This ensures that the buffers 0^t cannot be matched with large portions of any SZ codeword. We additionally leverage a key lemma (Lemma 9) which states that the edit distance between the codeword $C_{\text{in}}(i, 1 - y_i)$ and *any* substring of length less than $2t$ of the

uncorrupted codeword $C(x)$ has relative edit distance at least $\delta_{\text{in}}/2$. We use these two properties, along with key facts about the LCS matching, to yield an upper bound on the number of dangerous blocks, completing the analysis of our decoder.

Extension to relaxed locally correctable codes for insdel errors. Our construction also yields a strong Insdel Relaxed Locally Correctable Code (RLCC) with constant locality if the outer code is a weak Hamming RLCC. First, observe that bits of the codeword corresponding to the 0^t buffers are very easy to predict without even making any queries to the corrupted codeword. Thus, if we are asked to recover the j'th bit of the codeword and j corresponds to a buffer 0^t, we can simply return 0 without making any queries to the received word. Otherwise, if we are asked to recover the j'th bit of the codeword and j corresponds to block c_i, we can simulate the Hamming RLCC decoder (as above) on input i to obtain y_i (or $\perp$). Assuming that $y_i \in \{0,1\}$, we can compute the corresponding SZ encoding of (i, y_i) and obtain the original value of the block c_i and then recover the j'th bit of the original codeword. The analysis of the RLCC decoder is analogous to the RLDC decoder. See Section 6 for full details on both our weak Insdel RLDC and strong Insdel RLCC constructions.

Remark 1. The "adaptiveness" of our constructed Insdel RLDC/RLCC decoder is identical to that of the outer Hamming RLDC/RLCC decoder. In particular, the weak Hamming RLDC of Ben-Sasson et al. [BGH+06] has a non-adaptive decoder, making our final decoder non-adaptive as well. Similarly, we use a weak Hamming RLCC due to Asadi and Shinkar [AS21] for our Insdel LCC, which is also a non-adaptive decoder.

2 Open Questions

Exact "phase-transition" thresholds. Our results show that both in the Hamming and Insdel setting there is a constant q such that every q-query RLDC requires super-polynomial codeword length, while there exists a $(q+1)$-query RLDC of polynomial codeword length. Finding the precise q remains an intriguing open question. Further, a more refined understanding of codeword length for RLDCs making $3, 4, 5$ queries is another important question, which has lead to much progress in the understanding of the LDC variants.

Constant-query strong Insdel RLDCs/RLCCs. While we do construct the first weak RLDCs in the Insdel setting, the drawback of our constructions is the fact that our codes do not satisfy the third property of Definition 1. Building strong Insdel RLDCs remains an open question. We note that our lower bounds imply that for a constant number of queries, such codes (if they exist) must have exponential codeword length.

Applications of local Insdel codes. As previously mentioned, Hamming LDCs/RLDCs have so far found many applications such as private information retrieval, probabilistically checkable proofs, self-correction, fault-tolerant circuits, hardness amplification, and data structures. Are there analogous or new applications of the Insdel variants in the broader computing area?

Lower bounds for Hamming RLDCs/LDCs Our 2-query lower bound for Hamming RLDCs crucially uses the perfect completeness property of the decoder. An immediate question is whether the bound still holds if we allow the decoder to have imperfect completeness. We also note that the argument in our exponential lower bounds for 2-query Hamming RLDCs fail to hold for alphabets other than the binary alphabet, and we leave the extension to larger alphabet sizes as an open problem. Another related question is to understand if one can leverage perfect completeness and/or random restrictions to obtain improved lower bounds for $q \geq 3$-query standard Hamming LDCs. Perfect completeness has been explicitly used before to show exponential lower bounds for 2-query LCCs [BGT17].

2.1 Further discussion about related work

Insdel codes. The study of error correcting codes for insertions and deletions was initiated by Levenstein [Lev66]. While progress has been slow because constructing codes for insdel errors is strictly more

challenging than for Hamming errors, strong interest in these codes lately has led to many exciting results [SZ99, MK05, GW17, HS17, GL18, HSS18, HS18, BGZ18, CJLW18, CHL+19, CJLW19, GL19b, HRS19, Hae19, LTX19, GHS20, CGHL21, CL21] (See also the excellent surveys of [Slo02, MBT10, Mit08, HS21]).

Insdel LDCs. [OPS07] gave private-key constructions of LDCs with $m = \Theta(n)$ and locality polylog(n). [BKZ20] extended the construction from [OPS07] to settings where the sender/decoder do not share randomness, but the adversarial channel is resource bounded. [BB21] applied the [BBG+20] compiler to the private key Hamming LDC of [OPS07] (resp. resource bounded LDCs of [BKZ20]) to obtain private key Insdel LDCs (resp. resource bounded Insdel LDCs) with constant rate and polylog(n) locality.

Insdel LDCs have also been recently studied in *computationally bounded channels*, introduced in [Lip94]. Such channels can perform a bounded number of adversarial errors, but do not have unlimited computational power as the general Hamming channels. Instead, such channels operate with bounded resources. As expected, in many such limited-resource settings one can construct codes with strictly better parameters than what can be done generally [DGL04, MPSW05, GS16, SS16]. LDCs in these channels under Hamming error were studied in [OPS07, HO08, HOSW11, HOW15, BGGZ21, BKZ20]. [BB21] applied the [BBG+20] compiler to the Hamming LDC of [BKZ20] to obtain a constant rate Insdel LDCs with polylog(n) locality for resource bounded channels. The work of [CLZ20] proposes the notion of locally decodable codes with randomized encoding, in both the Hamming and edit distance regimes, and in the setting where the channel is oblivious to the encoded message, or the encoder and decoder share randomness. For edit error they obtain codes with $m = O(n)$ or $m = n \log n$ and polylog(n) query complexity. However, even in settings with shared randomness or where the channel is oblivious or resource bounded, there are no known constructions of Insdel LDCs with constant locality.

Locality in the study of insdel codes was also considered in [HS18], which constructs explicit synchronization strings that can be locally decoded.

2.2 Organization

The remainder of the paper is dedicated to proving all our results presented in Section 1. We give general preliminaries and recall some prior results used in our results in Section 3. We prove Theorem 1 in Section 4, prove Theorem 2 in Section 5, and prove Theorem 3 and Corollary 1 in Section 6.

3 Preliminaries

For natural number $n \in \mathbb{N}$, we let $[n] \coloneqq \{1, 2, \ldots, n\}$. We let "$\circ$" denote the standard string concatenation operation. For a string $x \in \{0,1\}^*$ of finite length, we let $|x|$ denote the length of x. For $i \in [|x|]$, we let $x[i]$ denote the i-th bit of x. Furthermore, for $I \subseteq [|x|]$, we let $x[I]$ denote the subsequence $x[i_1] \circ x[i_2] \circ \cdots \circ x[i_\ell]$, where $i_j \in I$ and $\ell = |I|$. For two strings $x, y \in \{0,1\}^n$ of length n, we let $\mathsf{HAM}(x, y)$ denote the *Hamming Distance* between x and y; i.e., $\mathsf{HAM}(x, y) \coloneqq \big|\{i \in [n] \colon x_i \neq y_i\}\big|$. Similarly, we let $\mathsf{ED}(x, y)$ denote the *Edit Distance* between x and y; i.e., $\mathsf{ED}(x, y)$ is the minimum number of insertions and deletions needed to transform string x into string y. We often discuss the *relative Hamming Distance* (resp., *relative Edit Distance*) between x and y, which is simply the Hamming Distance normalized by n, i.e., $\mathsf{HAM}(x, y)/n$ (resp., the Edit Distance normalized by $|x| + |y|$, i.e., $\mathsf{ED}(x, y)/(|x| + |y|)$). Finally, the *Hamming weight* of a string x is the number of non-zero entries of x, which we denote as $\mathsf{wt}(x) \coloneqq |\{i \in [|x|] \colon x_i \neq 0\}|$.

For completeness, we recall the definition of a classical locally decodable code, or just a *locally decodable code*.

Definition 2 (Locally Decodable Codes)**.** *A (q, δ, α)-Locally Decodable Code $C \colon \Sigma^n \to \Sigma^m$ is a code for which there exists a randomized decoder that makes at most q queries to the received word y and satisfies the following property: for every $i \in [n]$, if y is such that* dist$(y, C(x)) \leq \delta$ *for some unique $C(x)$, then the decoder, on input i, outputs x_i with probability $\geq \alpha$. Here, the randomness is taken over the random coins of the decoder, and* dist *is a normalized metric.*

If dist *is the relative Hamming distance, then we say that the code is a Hamming LDC; similarly, if* dist *is the relative edit distance, then we say that the code is an Insdel LDC.*

We recall the general 2-query Hamming LDC lower bound [KdW04, BRdW08].

Theorem 4 ([KdW04,BRdW08]). *For constants $\delta, \varepsilon \in (0,1/2)$ there exists a constant $c = c(\delta,\varepsilon) \in (0,1)$ such that if $C\colon \{0,1\}^n \to \{0,1\}^m$ is a $(2,\delta,1/2+\varepsilon)$ Hamming LDC then $m \geq 2^{cn-1}$.*

In our weak Insdel RLDC construction, we utilize a weak Hamming RLDC due to [BGH+06].

Lemma 1 ([BGH+06]). *For constants $\varepsilon, \delta \in (0,1/2)$ and $\gamma \in (0,1)$, there exists a constant $q = O_{\delta,\varepsilon}(1/\gamma^2)$ and a weak $(q,\delta,1/2+\varepsilon)$-Hamming RLDC $C\colon \{0,1\}^n \to \{0,1\}^m$ with $m = O(n^{1+\gamma})$. Moreover, the decoder of this code is non-adaptive.*

Our construction additionally utilizes the well-known Schulman-Zuckerman Insdel codes [SZ99].

Lemma 2 (Schulman-Zuckerman (SZ) Code [SZ99]). *There exists constants $\beta \geq 1$ and $\delta > 0$ such that for large enough values of $t > 0$, there exists a code $C\colon \{0,1\}^t \to \{0,1\}^{\beta t}$ capable of decoding from δ-fraction of Insdel errors and the additional property that for every $x \in \{0,1\}^t$ and $y = C(x)$, every substring y' of y with length at least 2 has Hamming weight $\geq \lfloor |y'|/2 \rfloor$.*

Our strong Insdel RLCC construction relies on a weak Hamming RLCC. We utilize the following weak Hamming RLCC implicit in [AS21].

Lemma 3 (Implied by Theorem 1 of [AS21]). *For every sufficiently large $q \in \mathbb{N}$ and $\varepsilon \in (0,1/2)$, there is a constant δ such that there exists a weak $(q,\delta,1/2+\varepsilon)$-relaxed Hamming Locally Correctable Code $C\colon \{0,1\}^n \to \{0,1\}^m$ with $m = n^{1+O(1/q)}$. Moreover, the decoder of this code is non-adaptive.*

4 Lower Bounds for 2-Query Hamming RLDCs

We prove Theorem 1 in this section. As a reminder, a weak (q,δ,α)-RLDC satisfies the first two conditions in Definition 1, and non-adaptive means the decoder makes queries according to a distribution which is independent of the received string y. Here we are interested in the case $q = 2$ and $\alpha = 1/2+\varepsilon$.

To avoid overloading first-time readers with heavy notations, we first present a proof of the lower bound for *non-adaptive* decoders, i.e., decoders with a query distribution independent of the received string. This proof will be easier to follow, while the crucial ideas behind it remain the same. The proof for the most general case is presented in the last subsection, with an emphasis on the nuances in dealing with adaptivity.

4.1 A Warm-up: the lower bound for non-adaptive decoders

In the following, we fix a relaxed decoder Dec for C. In this subsection, we assume that Dec is non-adaptive, and that it has the first two properties specified in Definition 1. To avoid technical details, we also assume Dec always makes exactly 2 queries (otherwise add dummy queries to make the query count exactly 2).

Given an index $i \in [n]$ and queries j,k made by $\mathsf{Dec}(i,\cdot)$, in the most general setting the output could be a random variable which depends on i and y_j, y_k, where y_j, y_k are the answers to queries j, k, respectively. An equivalent view is that the decoder picks a random function f according to some distribution and outputs $f(y_j,y_k)$. Let $\mathtt{DF}^i_{j,k}$ be the set of all decoding functions $f\colon \{0,1\}^2 \to \{0,1,\perp\}$ which are selected by $\mathsf{Dec}(i,\cdot)$ with non-zero probability when querying j,k. We partition the queries into the following two sets

$$F_i^0 := \left\{\{j,k\} \subseteq [m]\colon \forall f \in \mathtt{DF}^i_{j,k} \text{ the truth table of } f \text{ contains no } \text{“}\perp\text{”}\right\},$$
$$F_i^{\geq 1} := \left\{\{j,k\} \subseteq [m]\colon \exists f \in \mathtt{DF}^i_{j,k} \text{ the truth table of } f \text{ contains at least 1 } \text{“}\perp\text{”}\right\}.$$

Notation. Given a string $w \in \{0,1\}^m$ and a subset $S \subseteq [m]$, we denote $w[S] := (w_i)_{i\in S} \in \{0,1\}^{|S|}$. Given a Boolean function $f\colon \{0,1\}^n \to \{0,1\}$, and $\sigma \in \{0,1\}$, we write $f \restriction_{x_i=\sigma}$ to denote the restriction of f to the domain $\{\mathbf{x} \in \{0,1\}^n : x_i = \sigma\}$. For a sequence of restrictions, we simply write $f \restriction_{(x_{j_1},\ldots,x_{j_k})=(\sigma_1,\ldots,\sigma_k)}$, or $f_{J|\sigma}$ where $J = [n] \setminus \{j_1,\ldots,j_k\}$ and $\sigma = (\sigma_1,\ldots,\sigma_k)$. Note that $f_{J|\sigma}$ is a Boolean function over the domain $\{0,1\}^J$.

We will identify the encoding function of C as a collection of m Boolean functions

$$\mathcal{C} \coloneqq \left\{C_1, \ldots, C_m \colon \forall j \in [m], C_j \colon \{0,1\}^n \to \{0,1\}\right\}.$$

Namely, $C(x) = (C_1(x), C_2(x), \ldots, C_m(x))$ for all $x \in \{0,1\}^n$.

For $j \in [m]$, we say C_j is *fixable* by x_i if at least one of the restrictions $C_j \restriction_{x_i=0}$ and $C_j \restriction_{x_i=1}$ is a constant function. Denote

$$S_i \coloneqq \left\{j \in [m] \colon C_j \text{ is fixable by } x_i\right\}, \quad T_j \coloneqq \left\{i \in [n] \colon C_j \text{ is fixable by } x_i\right\},$$

and $w_j \coloneqq |T_j|$. Let

$$W \coloneqq \left\{j \in [m] \colon w_j \geq 3\ln(8/\delta)\right\}.$$

For $i \in [n]$ define the sets $S_{i,+} \coloneqq S_i \cap W$, and $S_{i,-} \coloneqq S_i \cap \overline{W}$.

Let $J \subseteq [n]$ and $\rho \in \{0,1\}^{\overline{J}}$. A code $C \colon \{0,1\}^n \to \{0,1\}^m$ restricted to $\mathbf{x}_{\overline{J}} = \rho$, denoted by $C_{J|\rho}$, is specified by the following collection of Boolean functions

$$\mathcal{C}_{J|\rho} \coloneqq \left\{C_j \restriction_{\mathbf{x}_{\overline{J}}=\rho} \colon j \in [m], C_j \restriction_{\mathbf{x}_{\overline{J}}=\rho} \text{ is not a constant function}\right\}.$$

Namely, we restrict each function C_j in $\mathcal{C}$ to $\mathbf{x}_{\overline{J}} = \rho$, and eliminate those that have become constant functions. $C_{J|\rho}$ encodes n'-bit messages into m'-bit codewords, where $n' = |J|$ and $m' = \left|\mathcal{C}_{J|\rho}\right| \leq m$.

We note that the local decoder Dec for C can also be used as a local decoder for $C_{J|\rho}$, while preserving all the parameters. This is because, Dec never needs to really read a codeword bit which has become a constant function under the restriction $J|\rho$.

The lemma below will be useful later in the proof. It shows that a constant fraction of the message bits can be fixed so that most codeword bits C_j with large w_j become constants.

Lemma 4. *There exist a set $J \subseteq [n]$ and assignments $\rho \in \{0,1\}^{\overline{J}}$ such that $|J| \geq n/6$, and $|W \setminus A| \leq \delta m/4$, where $A \subseteq W$ collects all codeword bits $j \in W$ such that $C_j \restriction_{\mathbf{x}_{\overline{J}}=\rho}$ is a constant function.*

Proof. Let J be a random subset formed by selecting each $i \in [n]$ independently with probability $1/3$. For each $j \in \overline{J}$, set $\rho_j = 0$ or $\rho_j = 1$ with probability $1/2$. We have $\mathbb{E}[|J|] = n/3$, and hence the Chernoff bound shows that $|J| < n/6$ with probability $\exp(-\Omega(n))$. Furthermore, for each $j \in W$, $C_j \restriction_{\mathbf{x}_{\overline{J}}=\rho}$ becomes a constant function except with probability $\delta/8$. This is because for each $i \in T_j$, $C_j \restriction_{x_i=0}$ or $C_j \restriction_{x_i=1}$ is a constant function, and either case happens with probability $1/3$. Therefore

$$\Pr\left[C_j \restriction_{\mathbf{x}_{\overline{J}}=\rho} \text{ is not constant}\right] \leq \left(1 - \frac{1}{3}\right)^{|T_j|} < e^{-|T_j|/3} \leq \frac{\delta}{8},$$

where the last inequality is due to $w_j = |T_j| \geq 3\ln(8/\delta)$, since $j \in W$.

By linearity of expectation and Markov's inequality, we have

$$\begin{aligned}
&\Pr\left[\sum_{j \in W} \mathbf{1}\left\{C_j \restriction_{\mathbf{x}_{\overline{J}}=\rho} \text{ is not constant}\right\} \geq \frac{\delta}{4}|W|\right] \\
\leq &\frac{\mathbb{E}\left[\sum_{j \in W} \mathbf{1}\left\{C_j \restriction_{\mathbf{x}_{\overline{J}}=\rho} \text{ is not constant}\right\}\right]}{\delta|W|/4} \\
= &\frac{\sum_{j \in W} \Pr\left[C_j \restriction_{\mathbf{x}_{\overline{J}}=\rho} \text{ is not constant}\right]}{\delta|W|/4} \\
\leq &\frac{\delta/8 \cdot |W|}{\delta|W|/4} \leq \frac{1}{2}.
\end{aligned}$$

Applying a union bound gives

$$\Pr\left[\left(|J| < n/6\right) \vee \left(\sum_{j\in W} \mathbf{1}\left\{C_j \restriction_{\mathbf{x}_{\overline{J}}=\rho} \text{ is not constant}\right\} \geq \frac{\delta}{4}|W|\right)\right] \leq \exp\left(-\Omega(n)\right) + \frac{1}{2} < 1.$$

Finally, we can conclude that there exist $J \subseteq [n]$ and $\rho \in \{0,1\}^{\overline{J}}$ such that $|J| \geq n/6$, and $C_j \restriction_{\mathbf{x}_{\overline{J}}=\rho}$ becomes a constant function for all but $\delta/4$ fraction of $j \in W$. □

Let $J \subseteq [n]$ and $\rho \in \{0,1\}^{\overline{J}}$ be given by the Lemma 4, and consider the restricted code $C_{J|\rho}$. By rearranging the codeword bits, we may assume $J = [n']$ where $n' = |J| \geq n/6$. Let $A \subseteq [m]$ be the set of codeword bits which get fixed to constants under $J|\rho$. We denote $W' \coloneqq W \setminus A$, $S'_i \coloneqq S_i \setminus A$, $S'_{i,-} \coloneqq S_{i,-} \setminus A$, and $S'_{i,+} \coloneqq S_{i,+} \setminus A$. Note that $|W'| = |W \setminus A| \leq \delta m/4$, and thus $|S'_{i,+}| = |S_{i,+} \cap W'| \leq \delta m/4$ for all $i \in [n']$. We emphasize that S'_i does not necessarily contain all codeword bits fixable by x_i in the restricted code $C_{J|\rho}$, as fixing some message bits may cause more codeword bits to be fixable by x_i.

We first show that the queries of C must have certain structures. The following claim characterizes the queries in $F_i^{\geq 1}$.

Claim 1. *Suppose $\{j,k\} \in F_i^{\geq 1}$. Then we must have $j,k \in S_i$.*

Proof. Let $\{j,k\} \in F_i^{\geq 1}$. Suppose for the sake of contradiction that $j \notin S_i$. This implies there are partial assignments $\sigma_{00}, \sigma_{01}, \sigma_{10}, \sigma_{11} \in \{0,1\}^{n-1}$ such that

$$C_j\left(\mathbf{x}_{-i} = \sigma_{00}, x_i = 0\right) = 0, \quad C_j\left(\mathbf{x}_{-i} = \sigma_{01}, x_i = 1\right) = 0,$$
$$C_j\left(\mathbf{x}_{-i} = \sigma_{10}, x_i = 0\right) = 1, \quad C_j\left(\mathbf{x}_{-i} = \sigma_{11}, x_i = 1\right) = 1,$$

where $\mathbf{x}_{-i}$ is defined as $\left(x_t : t \in [n] \setminus \{i\}\right)$.

Let $C_{00}, C_{01}, C_{10}, C_{11}$ be encodings of the corresponding assignments mentioned above. Since the relaxed decoder has perfect completeness, when $\mathsf{Dec}(i,\cdot)$ is given access to C_{00} or C_{10} it must output $x_i = 0$. Note that the j-th bit is different in C_{00} and C_{10}. Similarly, when $\mathsf{Dec}(i,\cdot)$ is given access to C_{01} or C_{11} it must output $x_i = 1$. However, this already takes up 4 entries in the truth table of any decoding function $f \in \mathtt{DF}^i_{j,k}$, leaving no space for any "$\perp$" entry. This contradicts with the assumption $\{j,k\} \in F_i^{\geq 1}$. □

Here is another way to view Claim 1 which will be useful later: Suppose $\{j,k\}$ is a query set such that $j \notin S_i$ (or $k \notin S_i$), then $\{j,k\} \in F_i^0$. In other words, conditioned on the event that some query is not contained in S_i, the decoder never outputs $\perp$.

The following claim characterizes the queries in F_i^0.

Claim 2. *Suppose $\{j,k\} \in F_i^0$, and $j \in S_i$. Then one of the following three cases occur: (1) $k \in S_i$, (2) $C_j = x_i$, or (3) $C_j = \neg x_i$.*

Proof. Since $j \in S_i$, we may, without loss of generality, assume that $C_j \restriction_{x_i=0}$ is a constant function. Let us further assume it is the constant-zero function. The proofs for the other cases are going to be similar.

Denote by $f(y_j, y_k)$ the function returned by $\mathsf{Dec}(i,\cdot)$ conditioned on reading $\{j,k\}$. Any function $f \in \mathtt{DF}^i_{j,k}$ takes values in $\{0,1\}$ since $\{j,k\} \in F_i^0$. Suppose case (1) does not occur, meaning that $C_k \restriction_{x_i=0}$ is not a constant function. Then there must be partial assignments $\sigma_{00}, \sigma_{01} \in \{0,1\}^{n-1}$ such that

$$C_k(x_i = 0, \mathbf{x}_{-i} = \sigma_{00}) = 0, \quad C_k(x_i = 0, \mathbf{x}_{-i} = \sigma_{01}) = 1.$$

Let C_{00} and C_{01} be the encodings of the corresponding assignments mentioned above. Due to perfect completeness of Dec, it must always output $x_i = 0$ when given access to C_{00} or C_{01}. That means $f(0,0) = f(0,1) = 0$.

Now we claim that $C_j \restriction_{x_i=1}$ must be the constant-one function. Otherwise there is a partial assignment $\sigma_{10} \in \{0,1\}^{n-1}$ such that

$$C_j(x_i = 1, \mathbf{x}_{-i} = \sigma_{10}) = 0.$$

Let C_{10} be the encoding of this assignment. On the one hand, due to perfect completeness $\mathsf{Dec}(i,\cdot)$ should always output $x_i = 1$ when given access to C_{10}. On the other hand, $\mathsf{Dec}(i,\cdot)$ outputs $f((C_{10})_j, 0) = f(0,0) = 0$. This contradiction shows that $C_j \restriction_{x_i=1}$ must be the constant-one function. Therefore $C_j = x_i$, i.e., case (2) occurs.

Similarly, when $C_j \restriction_{x_i=0}$ is the constant-one function, we can deduce that $C_j = \neg x_i$, i.e., case (3) occurs. □

We remark that Claim 1 and Claim 2 jointly show that for any query set $\{j,k\}$ made by $\mathsf{Dec}(i,\cdot)$ there are 2 essentially different cases: (1) both j,k lie inside S_i, and (2) both j,k lie outside S_i. The case $j \in S_i, k \notin S_i$ ($k \in S_i, j \notin S_i$, resp.) means that k (j, resp.) is a dummy query which is not used for decoding. Furthermore, conditioned on case (2), the decoder never outputs $\perp$.

Another important observation is that all properties of the decoder discussed above hold for the restricted code $C_{J|\rho}$, with S_i replaced by S'_i. This is because $C_{J|\rho}$ uses essentially the same decoder, except that it does not actually query any codeword bit which became a constant.

For a subset $S \subseteq [m]$, we say "$\mathsf{Dec}(i,\cdot)$ reads S" if the event "$j \in S$ and $k \in S$" occurs where $j,k \in [m]$ are the queries made by $\mathsf{Dec}(i,\cdot)$. The following lemma says that conditioned on $\mathsf{Dec}(i,\cdot)$ reads some subset S, there is a way of modifying the bits in S that flips the output of the decoder.

Lemma 5. *Let $S \subseteq [m]$ be a subset such that $\Pr[\mathsf{Dec}(i,\cdot) \text{ reads } S] > 0$. Then for any string $s \in \{0,1\}^m$ and any bit $b \in \{0,1\}$, there exists a string $z \in \{0,1\}^m$ such that $z[[m] \setminus S] = s[[m] \setminus S]$, and*

$$\Pr\left[\mathsf{Dec}(i,z) = 1-b \mid \mathsf{Dec}(i,\cdot) \text{ reads } S\right] = 1.$$

Proof. Let $x \in \{0,1\}^n$ be a string with $x_i = 1-b$. Let $z \in \{0,1\}^m$ be the string satisfying

$$z[S] = C(x)[S], \quad z[[m] \setminus S] = s[[m] \setminus S].$$

Since Dec has perfect completeness, we have

$$1 = \Pr\left[\mathsf{Dec}(i, C(x)) = x_i \mid \mathsf{Dec}(i,\cdot) \text{ reads } S\right] = \Pr\left[\mathsf{Dec}(i,z) = 1-b \mid \mathsf{Dec}(i,\cdot) \text{ reads } S\right].$$

□

The next lemma is a key step in our proof. It roughly says that there is a local decoder for x_i in the standard sense as long as the size of S_i is not too large.

Lemma 6. *Suppose $i \in [n]$ is such that $|S_i| \le \delta m/2$. Then there is a $(2, \delta/2, 1/2+\varepsilon)$-local decoder D_i for i. In other words, for any $x \in \{0,1\}^n$ and $y \in \{0,1\}^m$ such that $\mathsf{HAM}(C(x), y) \le \delta m/2$, we have*

$$\Pr\left[D_i(y) = x_i\right] \ge \frac{1}{2} + \varepsilon,$$

and D_i makes at most 2 queries into y.

Proof. Let $i \in [n]$ be such that $|S_i| \le \delta m/2$. The local decoder D_i works as follows. Given $x \in \{0,1\}^n$ and $y \in \{0,1\}^m$ such that $\mathsf{HAM}(C(x), y) \le \delta m/2$, D_i obtains a query set Q according to the query distribution of $\mathsf{Dec}(i,\cdot)$ conditioned on $Q \subseteq [m] \setminus S_i$. Then D_i finishes by outputting the result returned by $\mathsf{Dec}(i,\cdot)$.

Denote by E_i the event "$\mathsf{Dec}(i,\cdot)$ reads $[m] \setminus S_i$", i.e., both two queries made by $\mathsf{Dec}(i,\cdot)$ lie outside S_i. In order for the conditional distribution to be well-defined, we need to argue that E_i occurs with non-zero probability. Suppose this is not the case, meaning that $Q \cap S_i \neq \varnothing$ for all possible query set Q. Let $z \in \{0,1\}^m$ be the string obtained by applying Lemma 5 with $S = S_i$, $s = C(x)$ and $b = x_i$. Claim 1 and Claim 2 jointly show that either $Q \subseteq S_i$, or the decoder's output does not depend on the answers to queries in $Q \setminus S_i$. In any case, the output of $\mathsf{Dec}(i,z)$ depends only on $z[S_i]$. However, by the choice of z we now have a contradiction since

$$\frac{1}{2} + \varepsilon \le \Pr\left[\mathsf{Dec}(i,z) \in \{x_i, \perp\}\right] = \Pr\left[\mathsf{Dec}(i,z) \in \{x_i, \perp\} \mid \mathsf{Dec}(i,\cdot) \text{ reads } S_i\right] = 0,$$

where the first inequality is due to $\mathsf{HAM}(C(x), z) \le |S_i| < \delta m$ and the relaxed decoding property of Dec.

By definition of D_i, it makes at most 2 queries into y. Its success rate is given by

$$\Pr[D_i(y) = x_i] = \Pr[\mathsf{Dec}(i, y) = x_i \mid E_i].$$

Therefore it remains to show that

$$\Pr\left[\mathsf{Dec}(i, y) = x_i \mid E_i\right] \geq \frac{1}{2} + \varepsilon.$$

Let z be the string obtained by applying Lemma 5 with $S = S_i$, $s = y$ and $b = x_i$. From previous discussions we see that conditioned on $\overline{E_i}$ (i.e., the event E_i does not occur), the output of $\mathsf{Dec}(i, z)$ only depends on $z[S_i]$. Therefore

$$\Pr\left[\mathsf{Dec}(i, z) \in \{x_i, \perp\} \mid \overline{E_i}\right] = 1 - \Pr\left[\mathsf{Dec}(i, z) = 1 - x_i \mid \overline{E_i}\right] = 0. \tag{1}$$

We also have that z is close to $C(x)$ since

$$\mathsf{HAM}(z, C(x)) \leq \mathsf{HAM}(z, y) + \mathsf{HAM}(y, C(x)) \leq |S_i| + \delta m/2 \leq \delta m.$$

Thus, the relaxed decoding property of Dec gives

$$\Pr\left[\mathsf{Dec}(i, z) \in \{x_i, \perp\}\right] \geq \frac{1}{2} + \varepsilon.$$

On the other hand, we also have

$$\begin{aligned}
&\Pr\left[\mathsf{Dec}(i, z) \in \{x_i, \perp\}\right] \\
=&\Pr\left[\mathsf{Dec}(i, z) \in \{x_i, \perp\} \mid \overline{E_i}\right] \cdot \Pr\left[\overline{E_i}\right] + \Pr\left[\mathsf{Dec}(i, z) \in \{x_i, \perp\} \mid E_i\right] \cdot \Pr\left[E_i\right] \\
=&\Pr\left[\mathsf{Dec}(i, z) \in \{x_i, \perp\} \mid \overline{E_i}\right] \cdot \Pr\left[\overline{E_i}\right] + \Pr\left[\mathsf{Dec}(i, y) \in \{x_i, \perp\} \mid E_i\right] \cdot \Pr\left[E_i\right] \\
& \qquad\qquad (z[[m] \setminus S_i] = y[[m] \setminus S_i]) \\
=&\Pr\left[\mathsf{Dec}(i, y) \in \{x_i, \perp\} \mid E_i\right] \cdot \Pr\left[E_i\right] \qquad \text{Equation (1)} \\
\leq&\Pr\left[\mathsf{Dec}(i, y) \in \{x_i, \perp\} \mid E_i\right].
\end{aligned}$$

Note that by Claim 1, conditioned on E_i, $\mathsf{Dec}(i, \cdot)$ never outputs "$\perp$". We thus have

$$\Pr\left[\mathsf{Dec}(i, y) = x_i \mid E_i\right] \geq \frac{1}{2} + \varepsilon.$$

□

We remark once again that the above lemma holds for the restricted code $C_{J|\rho}$, with S_i replaced by S'_i.

Below we prove an exponential lower bound for non-adaptive 2-query Hamming RLDCs.

Proposition 1. *Let $C\colon \{0,1\}^n \to \{0,1\}^m$ be a non-adaptive weak $(2, \delta, 1/2 + \varepsilon)$-RLDC. Then $m = 2^{\Omega_{\delta,\varepsilon}(n)}$.*

Proof. Let $C_{J|\rho}\colon \{0,1\}^{n'} \to \{0,1\}^{m'}$ be the restricted code where $J|\rho$ is given by Lemma 4, and $A \subseteq [m]$ be the set of codeword bits which get fixed to constants. We also let $S'_i := S_i \setminus A$, $S'_{i,-} = S_{i,-} \setminus A$, $S'_{i,+} = S_{i,+} \setminus A$.

Denote $T'_j := \left\{i \in [n']\colon j \in S'_i\right\}$. Since $S'_i \subseteq S_i$ for each i, we also have $T'_j \subseteq T_j$ for each j. In particular, for each $j \notin W' \subseteq W$, we have $|T'_j| \leq |T_j| \leq 3\ln(8/\delta)$. Therefore

$$\mathop{\mathbb{E}}_{i \in [n']}\left[\left|S'_{i,-}\right|\right] = \frac{1}{n'}\sum_{i=1}^{n'} |S'_{i,-}| = \frac{1}{n'}\sum_{j \in [m'] \setminus W'} |T'_j| \leq 3\ln(8/\delta) \cdot \frac{m'}{n'}.$$

Therefore by Markov's inequality,

$$\Pr_{i \in [n']}\left[|S'_{i,-}| > \delta m'/4\right] \leq \frac{12\ln(8/\delta)}{\delta n'} = O_\delta\left(\frac{1}{n'}\right).$$

In other words, there exists $I \subseteq [n']$ of size $|I| \geq n' - O_\delta(1)$ such that $|S'_{i,-}| \leq \delta m'/4$ for all $i \in I$. For any such $i \in I$, we have $|S'_i| = |S'_{i,-}| + |S'_{i,+}| \leq \delta m'/4 + \delta m'/4 = \delta m'/2$. By Lemma 6, we can view $C_{J|\rho}$ as a $(2, \delta/2, 1/2 + \varepsilon)$-LDC for message bits in I (for instance, we can arbitrarily fix the message bits outside I), where $|I| > n' - O_\delta(1) = \Omega(n)$. Finally, the statement of the proposition follows from Theorem 4. □

4.2 Lower bounds for adaptive 2-Query Hamming RLDCs

Now we turn to the actual proof, which still works for possibly adaptive decoders. Let C be a weak $(2, \delta, 1/2 + \varepsilon)$-RLDC with perfect completeness. We fix a relaxed decoder Dec for C. Without loss of generality, we assume Dec works as follows: on input $i \in [n]$, $\mathsf{Dec}(i, \cdot)$ picks the first query $j \in [m]$ according to a distribution $\mathcal{D}_i$. Let $b \in \{0,1\}$ be the answer to this query. Then Dec picks the second query $k \in [m]$ according to a distribution $\mathcal{D}_{i;j,b}$, and obtains an answer $b' \in \{0,1\}$. Finally, Dec outputs a random variable $X_{i;j,b,k,b'} \in \{0, 1, \perp\}$.

We partition the support of $\mathcal{D}_i$ into the following two sets:

$$F_i^0 \coloneqq \left\{ j \in \mathrm{supp}(\mathcal{D}_i) \colon \forall b, b' \in \{0,1\}, k \in \mathrm{supp}(\mathcal{D}_{i;j,b,k,b'}), \Pr[X_{i;j,b,k,b'} = \perp] = 0 \right\},$$
$$F_i^{>0} \coloneqq \left\{ j \in \mathrm{supp}(\mathcal{D}_i) \colon \exists b, b' \in \{0,1\}, k \in \mathrm{supp}(\mathcal{D}_{i;j,b,k,b'}), \Pr[X_{i;j,b,k,b'} = \perp] > 0 \right\}.$$

We will still apply the restriction guaranteed by Lemma 4 to C. The sets S_i, T_j, W, $S_{i,-}$, $S_{i,+}$ (are their counterparts for $C_{J|\rho}$) are defined in the exact same way.

The following claim is adapted from Claim 1.

Claim 3. $(\mathrm{supp}(\mathcal{D}_i) \setminus S_i) \subseteq F_i^0$.

Proof. Let $j \in \mathrm{supp}(\mathcal{D}_i) \setminus S_i$ and we will show $j \in F_i^0$. By the definition of S_i, $j \notin S_i$ means that there are partial assignments $\sigma_{00}, \sigma_{01}, \sigma_{10}, \sigma_{11} \in \{0,1\}^{n-1}$ such that

$$C_j(\mathbf{x}_{-i} = \sigma_{00}, x_i = 0) = 0, \quad C_j(\mathbf{x}_{-i} = \sigma_{01}, x_i = 1) = 0,$$
$$C_j(\mathbf{x}_{-i} = \sigma_{10}, x_i = 0) = 1, \quad C_j(\mathbf{x}_{-i} = \sigma_{11}, x_i = 1) = 1,$$

where $\mathbf{x}_{-i}$ is defined as $\left(x_t \colon t \in [n] \setminus \{i\}\right)$.

Let $C_{00}, C_{01}, C_{10}, C_{11}$ be encodings of the corresponding assignments mentioned above. Consider an arbitrary query $k \in \mathrm{supp}(\mathcal{D}_{i;j,0})$, and let b'_1, b'_2 be the k-th bit of C_{00} and C_{01}, respectively. We note that $X_{i;j,0,k,b'_1}$ is the output of $\mathsf{Dec}(i, C_{00})$ conditioned on the queries j, k, and $X_{i;j,0,k,b'_2}$ is the output of $\mathsf{Dec}(i, C_{01})$ conditioned on the queries j, k. Due to perfect completeness of Dec, we have

$$\Pr[X_{i;j,0,k,b'_1} = 0] = 1, \quad \Pr[X_{i;j,0,k,b'_2} = 1] = 1.$$

Therefore, it must be the case that $b'_1 \neq b'_2$, which implies that $\Pr[X_{i;j,0,k,b'} = \perp] = 0$ for any $b' \in \{0,1\}$.

An identical argument shows that $\Pr[X_{i;j,1,k,b'} = \perp] = 0$ for any $k \in \mathrm{supp}(\mathcal{D}_{i;j,1})$ and $b' \in \{0,1\}$. Thus we have shown $j \in F_i^0$. □

We remark that the above claim also implies $F_i^{>0} \subseteq S_i$, since $\mathrm{supp}(\mathcal{D}_i)$ is a disjoint union of F_i^0 and $F_i^{>0}$. In other words, conditioned on the event that the first query j is not contained in S_i, the decoder never outputs $\perp$.

The next claim is adapted from Claim 2.

Claim 4. *Let $j \in \mathrm{supp}(\mathcal{D}_i) \cap S_i$. For any $b \in \{0,1\}$ one of the following three cases occurs:*

1. $\mathrm{supp}(\mathcal{D}_{i;j,b}) \subseteq S_i$;
2. *For any $k \in \mathrm{supp}(\mathcal{D}_{i;j,b}) \setminus S_i$, $\Pr[X_{i;j,b,k,0} = b] = \Pr[X_{i;j,b,k,1} = b] = 1$;*
3. *For any $k \in \mathrm{supp}(\mathcal{D}_{i;j,b}) \setminus S_i$, $\Pr[X_{i;j,b,k,0} = 1 - b] = \Pr[X_{i;j,b,k,1} = 1 - b] = 1$.*

Proof. Since $j \in S_i$, we may, without loss of generality, assume that $C_j \restriction_{x_i = 0}$ is a constant function. Let us further assume $C_j \restriction_{x_i = 0} \equiv 0$. The proofs for the other cases are going to be similar.

Suppose $\mathrm{supp}(\mathcal{D}_{i;j,0}) \not\subseteq S_i$, and let $k \in \mathrm{supp}(\mathcal{D}_{i;j,0}) \setminus S_i$. By the definition of S_i, $k \notin S_i$ means that there are partial assignments $\sigma_{00}, \sigma_{01} \in \{0,1\}^{n-1}$ such that

$$C_k(x_i = 0, \mathbf{x}_{-i} = \sigma_{00}) = 0, \quad C_k(x_i = 0, \mathbf{x}_{-i} = \sigma_{01}) = 1.$$

Let C_{00} and C_{01} be the encodings of the corresponding assignments mentioned above. We note that $X_{i;j,0,k,0}$ and $X_{i;j,0,k,1}$ are the outputs of $\mathsf{Dec}(i, C_{00})$ and $\mathsf{Dec}(i, C_{01})$, respectively, conditioned on the queries j, k. Due to perfect completeness of Dec, we must have

$$\Pr[X_{i;j,0,k,0} = 0] = \Pr[X_{i;j,0,k,1} = 0] = 1,$$

since both C_{00} and C_{01} encode messages with $x_i = 0$.

Now we claim that $C_j \restriction_{x_i=1} \equiv 1$ must hold. Otherwise there is a partial assignment $\sigma_{10} \in \{0,1\}^{n-1}$ such that

$$C_j(x_i = 1, \mathbf{x}_{-i} = \sigma_{10}) = 0.$$

Let C_{10} be the encoding of this assignment, and let $b' \in \{0,1\}$ be the k-th bit of C_{10}. On the one hand, $X_{i;j,0,k,b'}$ is the output $\mathsf{Dec}(i, C_{10})$ conditioned on the queries j, k, and we have just established

$$\Pr[X_{i;j,0,k,b'} = 0] = 1.$$

On the other hand, $\mathsf{Dec}(i, C_{10})$ should output $x_i = 1$ with probability 1 due to perfect completeness. This contradiction shows that $C_j \restriction_{x_i=1} \equiv 1$.

Similarly, suppose $\mathrm{supp}(\mathcal{D}_{i;j,1}) \not\subseteq S_i$ and let $k \in \mathrm{supp}(\mathcal{D}_{i;j,1}) \setminus S_i$, meaning that there are partial assignments $\sigma_{10}, \sigma_{11} \in \{0,1\}^{n-1}$ such that

$$C_k(x_i = 1, \mathbf{x}_{-i} = \sigma_{10}) = 0, \quad C_k(x_i = 1, \mathbf{x}_{-i} = \sigma_{11}) = 1.$$

Let C_{10} and C_{11} be the corresponding encodings, and note that $X_{i;j,1,k,0}$ and $X_{i;j,1,k,1}$ are the outputs of $\mathsf{Dec}(i, C_{10})$ and $\mathsf{Dec}(i, C_{11})$, respectively, conditioned on the queries j, k. Perfect completeness of Dec implies

$$\Pr[X_{i;j,1,k,0} = 1] = \Pr[X_{i;j,1,k,1} = 1] = 1,$$

since both C_{10} and C_{11} encode messages with $x_i = 1$.

So far we have shown that for any $b \in \{0,1\}$ such that $\mathrm{supp}(\mathcal{D}_{i;j,b}) \not\subseteq S_i$, it holds that

$$\forall k \in \mathrm{supp}(\mathcal{D}_{i;j,b}) \setminus S_i, \quad \Pr[X_{i;j,b,k,0} = b] = \Pr[X_{i;j,b,k,1} = b] = 1,$$

provided that $C_j \restriction_{x_i=0} \equiv 0$. In case of $C_j \restriction_{x_i=0} \equiv 1$, we can use an identical argument to deduce that for any $b \in \{0,1\}$ such that $\mathrm{supp}(\mathcal{D}_{i;j,b}) \not\subseteq S_i$, it holds that

$$\forall k \in \mathrm{supp}(\mathcal{D}_{i;j,b}) \setminus S_i, \quad \Pr[X_{i;j,b,k,0} = 1 - b] = \Pr[X_{i;j,b,k,1} = 1 - b] = 1.$$

□

Here is another way to view Claim 4: conditioned on the event that the first query j is contained in S_i, either the second query k is also contained in S_i, or the output $X_{i;j,b,k,b'}$ is independent of the answer b' to query k. In either case, the decoder's output depends solely on the S_i-portion of the received string.

Once again, the conclusions of Claim 3 and Claim 4 hold for $C_{J|\rho}$, with S_i replaced by S'_i.

Finally, we are ready to prove Theorem 1. We recall the Theorem below.

Theorem 1. *Let $C\colon \{0,1\}^n \to \{0,1\}^m$ be a weak adaptive $(2, \delta, 1/2 + \varepsilon)$-RLDC. Then $m = 2^{\Omega_{\delta,\varepsilon}(n)}$.*

Proof. The proof is almost identical to the one for Proposition 1. First, we can show that there exists $I \subseteq [n']$ of size $|I| \geq n' - O_\delta(1) = \Omega(n)$ such that $|S'_{i,-}| \leq \delta m/4$ for all $i \in I$, and hence $|S'_i| = |S'_{i,-}| + |S'_{i,+}| \leq \delta m/2$. Second, similar to the proof of Lemma 6, for each $i \in I$ we can construct a decoder D_i for x_i as follows. D_i restarts $\mathsf{Dec}(i, \cdot)$ until it makes a first query $j \in [m'] \setminus S'_i$. Then D_i finishes simulating $\mathsf{Dec}(i, \cdot)$ and returns its output. With the help of Claim 3 and Claim 4, the same analysis in Lemma 6 shows that D_i never returns $\perp$, and that the probability of returning x_i is at least $1/2 + \varepsilon$. Finally, the theorem follows from Theorem 4. □

5 Lower Bounds for Strong Insdel RLDCs

In this section, we prove Theorem 2. We remind the readers that a strong $(q, \delta, \alpha, \rho)$-Insdel RLDC satisfies all 3 conditions in Definition 1, and here we are mainly interested in the case where q is a constant and $\alpha = 1/2 + \beta$ for $\beta > 0$. In fact, Theorem 2 would still hold even without perfect completeness (i.e., Condition 1 in Definition 1), as our proof does not rely on this condition. A corollary to this observation is that essentially the same lower bound also holds for strong Insdel RLDCs with adaptive decoders. This is because the Katz-Trevison reduction from adaptive to non-adaptive decoders does preserve Condition 2 and 3 in Definition 1, with the same ρ and mildly worse $\alpha = 1/2 + \beta/2^{q-1}$ (see Footnote 1).

Our proof relies on the following result, which is implicit in [BCG+22]. [BCG+22] shows an exponential lower bound on the length of constant-query Insdel LDCs. The core of their argument is the construction of an error distribution $\mathcal{D}$, whereby they derive necessary properties of the code to imply the exponential lower bound. As remarked in Section 4.1 of [BCG+22], $\mathcal{D}$ is oblivious to the decoding algorithm. That means their result holds even if the code is required to handle an error pattern much more innocuous than adversarial errors. This stronger statement allows us to define the notion of "locally decodable on average against $\mathcal{D}$", which would otherwise not be well-defined if the adversary is adaptive to the decoding strategy.

Theorem 5 ([BCG+22])**.** *Let $\delta \in (0,1)$ be a constant. There exists a channel $\mathfrak{D}$ for m-bit strings with the following properties.*

- *For every $s \in \{0,1\}^m$, $\Pr_{s' \sim \mathfrak{D}(s)}[\mathsf{ED}(s', s) > \delta \cdot 2m] < \mathsf{negl}(m)$.*
- *Suppose $C\colon \{0,1\}^n \to \{0,1\}^m$ is a code which is locally decodable on average against $\mathfrak{D}$. Formally, there is a randomized algorithm Dec satisfying*

$$\forall i \in [n], \quad \Pr_{\substack{x \in \{0,1\}^n \\ y \sim \mathfrak{D}(C(x))}} \left[\mathsf{Dec}(i, y) = x_i\right] \geq \frac{1}{2} + \varepsilon,$$

where the probability is taken over the uniform random choice of $x \in \{0,1\}^n$, the randomness of $\mathfrak{D}$, and the randomness of Dec. Furthermore, Dec makes at most q non-adaptive queries into y in each invocation. Then for every $q \geq 2$ there is a constant $\kappa = \kappa(q, \delta, \varepsilon)$ such that

$$m = \exp\left(\kappa \cdot n^{1/(2q-3)}\right).$$

As a side note, in the definition of Insdel LDCs in [BCG+22], the decoder also has the length of the received string y as an input. However, the channel $\mathfrak{D}$ constructed in [BCG+22] ensures that $|y| = m$ except with exponentially small probability, where $y \sim \mathfrak{D}(C(x))$. For this reason, we omit this extra input as it almost gives no information to the decoder.

The proof of Theorem 2 consists of two steps, and they are captured by Lemma 7 and Lemma 8 below. The first step is a straightforward confidence amplification step which boosts the success rate α by running the decoding algorithm multiple times. In the second step, we show that if α is sufficiently close to 1, a strong Insdel RLDC will imply a standard Insdel LDC that is decodable on average against the channel $\mathfrak{D}$ mentioned in Theorem 5, which is sufficient for deriving the exponential lower bound.

Lemma 7. *Let $C\colon \{0,1\}^n \to \{0,1\}^m$ be a non-adaptive strong $(q, \delta, 1/2 + \beta, \rho)$-Insdel RLDC where $\beta > 0$. Then for any $\varepsilon > 0$, C is also a non-adaptive strong $(q \cdot \ln(1/\varepsilon)/(2\beta^2), \delta, 1 - \varepsilon, \rho)$-Insdel RLDC.*

Proof. Let Dec be a relaxed local decoder for C. For some integer T to be decided, consider the following alternative local decoder Dec_T for C. On input (y, m, i), Dec_T independently runs $\mathsf{Dec}(y, m, i)$ for T times, and obtains outputs $r_1, r_2, \ldots, r_T \in \{0, 1, \perp\}$. For $b \in \{0, 1, \perp\}$ we denote

$$S_b \coloneqq \left\{t \in [T]\colon r_t = b\right\}.$$

Dec_T outputs 0 or 1 if $|S_0| \geq T/2$ or $|S_1| \geq T/2$, respectively. Otherwise Dec_T outputs $\perp$. Clearly, Dec_T is non-adaptive if Dec is non-adaptive.

Now we prove the three properties of Dec_T. Perfect completeness is easy to see. The relaxed decoding property is violated when $|S_{1-x_i}| \geq T/2$. By the Chernoff bound, this happens with probability at most $e^{-2\beta^2 T}$, since for each $t \in [T]$ we have

$$\Pr[r_t = 1 - x_i] = 1 - \Pr[r_t \in \{x_i, \perp\}] \leq 1 - \left(\frac{1}{2} + \beta\right) = \frac{1}{2} - \beta.$$

Let $I_y \subseteq [n]$ be the subset given by the third property of Dec. That is, for each $i \in I_y$ and $t \in [T]$, we have

$$\Pr[r_t = x_i] \geq \frac{1}{2} + \beta.$$

Again, by the Chernoff bound we have $\Pr[|S_{x_i}| < T/2] < e^{-2\beta^2 T}$, for each $i \in I_y$.

Finally, we take $T = \ln(1/\varepsilon)/(2\beta^2)$ which ensures $e^{-2\beta^2 T} \leq \varepsilon$. We note that Dec_T makes $q \cdot T = q \cdot \ln(1/\varepsilon)/(2\beta^2)$ queries to y. □

Lemma 8. *Let $C\colon \{0,1\}^n \to \{0,1\}^m$ be a non-adaptive $(q, \delta, \alpha, \rho)$-Insdel RLDC. Suppose $\rho\alpha + (1-\rho)\alpha/2 = 1/2 + \varepsilon$ for some $\varepsilon > 0$. Then there exists a non-adaptive decoder Dec and a subset $I \subseteq [n]$ of size at least $\varepsilon n/2$ such that for every $i \in I$, we have*

$$\Pr_{\substack{x \in \{0,1\}^n \\ y \sim \mathfrak{D}(C(x))}} \left[\mathsf{Dec}(i, y) = x_i\right] \geq \frac{1}{2} + \frac{\varepsilon}{4}.$$

The probability is taken over the uniform random choice of $x \in \{0,1\}^n$, the randomness of $\mathfrak{D}$, and the randomness of Dec.

Proof. Let Dec_0 be the relaxed decoder for C. The local decoder Dec will simulate Dec_0 and output the result, except when Dec_0 returns "$\perp$", Dec instead returns a uniform random bit. Clearly, Dec is non-adaptive if Dec_0 is non-adaptive.

We note that $\mathfrak{D}$ introduces at most $\delta \cdot 2m$ insertions and deletions except with probability $\mathsf{negl}(m)$. Therefore by definition of strong Insdel RLDCs (specifically Condition 2 and 3 in Definition 1), for a random index $i \in [n]$, we have

$$\Pr_{\substack{i \in [n] \\ x \in \{0,1\}^n \\ y \sim \mathfrak{D}(C(x))}} \left[\mathsf{Dec}(i, y) = x_i\right] \geq \rho\alpha + (1-\rho) \cdot \frac{\alpha}{2} - \mathsf{negl}(m) \geq \frac{1}{2} + \frac{\varepsilon}{2},$$

for large enough m (and thus n). The first inequality is because conditioned on $i \in I_y$ (which happens with probability $\geq \rho$ for any y), $\mathsf{Dec}(i, y) = x_i$ with probability α; and conditioned on $i \notin I_y$ (which happens with probability $\leq 1 - \rho$), the random guess provides an additional success rate of $\alpha/2$. We will show that the following subset of indices has large density:

$$I = \left\{ i \in [n] \colon \Pr_{\substack{x \in \{0,1\}^n \\ y \sim \mathfrak{D}(C(x))}} \left[\mathsf{Dec}(i, y) = x_i\right] \geq \frac{1}{2} + \frac{\varepsilon}{4} \right\}.$$

Denote by $p = |I|/n$ the density of I. We have

$$\begin{aligned}
\frac{1}{2} + \frac{\varepsilon}{2} &\leq \Pr_{\substack{i \in [n] \\ x \in \{0,1\}^n \\ y \sim \mathfrak{D}(C(x))}} \left[\mathsf{Dec}(i, y) = x_i\right] \\
&= \Pr[i \in I] \cdot \Pr_{\substack{x \in \{0,1\}^n \\ y \sim \mathfrak{D}(C(x))}} \left[\mathsf{Dec}(i, y) = x_i \mid i \in I\right] + \Pr[i \notin I] \cdot \Pr_{\substack{x \in \{0,1\}^n \\ y \sim \mathfrak{D}(C(x))}} \left[\mathsf{Dec}(i, y) = x_i \mid i \notin I\right] \\
&\leq p \cdot 1 + (1-p) \cdot \left(\frac{1}{2} + \frac{\varepsilon}{4}\right) \\
&\leq \frac{1}{2} + \frac{\varepsilon}{4} + \frac{p}{2}.
\end{aligned}$$

It follows that $p \geq \varepsilon/2$. □

Now we are ready to prove Theorem 2. We recall the theorem below.

Theorem 2. *Let $C\colon \{0,1\}^n \to \{0,1\}^m$ be a non-adaptive strong $(q,\delta,1/2+\beta,\rho)$-Insdel RLDC where $\beta > 0$. Then for every $q \geq 2$ there is a constant $c_1 = c_1(q,\delta,\beta,\rho)$ such that*

$$m = \exp\left(c_1 \cdot n^{\Omega_\rho(\beta^2/q)}\right).$$

Furthermore, the same bound holds even if C does not have perfect completeness. If C has an adaptive decoder, the same bound holds with β replaced by $\beta/2^{q-1}$. Formally, there exists a constant $c_2 = c_1(q,\delta,\beta/2^{q-1},\rho)$ such that

$$m = \exp\left(c_2 \cdot n^{\Omega_\rho(\beta^2/(q2^{2q}))}\right).$$

Proof. We first prove the bound for non-adaptive decoders. Taking $\varepsilon = \rho/4$ in Lemma 7, we have that C is also a non-adaptive strong (q',δ,α',ρ)-Insdel RLDC, where $q' = q\cdot\ln(4/\rho)/(2\beta^2)$ and $\alpha' = 1-\rho/4$. Note that

$$\rho\alpha' + \frac{(1-\rho)\alpha'}{2} = \frac{(1+\rho)\alpha'}{2} = \frac{(1+\rho)(1-\rho/4)}{2} \geq \frac{1}{2} + \frac{\rho}{4}.$$

Thus we can apply Lemma 8 with $\varepsilon = \rho/4$ to obtain a subset $I \subseteq [n]$ of size $|I| \geq \rho n/8$, and a decoder Dec satisfying

$$\Pr_{\substack{x\in\{0,1\}^n \\ y\sim\mathfrak{D}(C(x))}}\left[\mathsf{Dec}(i,y) = x_i\right] \geq \frac{1}{2} + \varepsilon' := \frac{1}{2} + \frac{\rho}{16}$$

for every $i \in I$. By Theorem 5, for some constant $c_1 = c_1(q,\delta,\beta,\rho)$ we have

$$m \geq \exp\left(\kappa(q',\delta,\varepsilon')\cdot|I|^{1/(2q'-3)}\right) \geq \exp\left(c_1\cdot n^{1/(2q')}\right) = \exp\left(c_1\cdot n^{\beta^2/(q\cdot\ln(4/\rho))}\right)$$

as desired.

For the adaptive case, we note that the proof does not rely on perfect completeness of the decoder (i.e., Condition 1 in Definition 1). Therefore, we can apply the Katz-Trevisan reduction (see Footnote 1) to obtain a non-adaptive decoder for C which satisfies Condition 2 and 3 in Definition 1, with the same ρ and mildly worse $\alpha = 1/2 + \beta/2^{q-1}$. The proof argument presented in this section still applies to such a non-adaptive decoder, whereby we can derive the same lower bound except with β replaced by $\beta/2^{q-1}$. □

We end this section with a remark on linear/affine weak 2-query insdel RLDCs. In Section 4, a transformation from RLDCs to standard LDCs was given for Hamming errors. This is done by fixing some message bits to 0 or 1, together with other modifications to the decoding algorithm. Here the code will remain affine if the initial code is linear or affine. One key step in the analysis is using the perfect completeness condition to deduce structural properties about the queries. We note that the same argument would yield the same query structure for weak insdel RLDCs. Altogether, this allows us to use the impossibility result for affine 2-query insdel LDCs [BCG+22] to conclude the following.

Corollary 2. *For any linear or affine weak $(2,\delta,\varepsilon)$ insdel RLDC $C\colon \{0,1\}^n \to \{0,1\}^m$, we have $n = O_{\delta,\varepsilon}(1)$.*

6 Weak Insdel RLDC and Strong Insdel RLCC Constructions

We prove Theorem 3 and Corollary 1 in this section. As a reminder, a weak (q,δ,α)-Insdel RLDC satisfies the first two conditions of Definition 1. Our constructions will have constant locality q, constant error-tolerance δ, and codeword length $m = O(n^{1+\gamma})$ for any $\gamma \in (0,1)$.

Algorithm 1: Encoding algorithm C.

Input : A message $x \in \{0,1\}^n$.
Output : A codeword $c \in \{0,1\}^m$.
Hardcoded : An outer encoder $C_{\text{out}} \colon \{0,1\}^n \to \{0,1\}^k$ and an inner encoder $C_{\text{in}} \colon [k] \times \{0,1\} \to \{0,1\}^t$.

Compute $y \coloneqq y_1 \circ y_2 \circ \cdots \circ y_k = C_{\text{out}}(x)$.
foreach $j \in [k]$ **do**
  Compute $c_j \coloneqq C_{\text{in}}(j, y_j)$.
Define $c \coloneqq c_1 \circ 0^t \circ c_2 \circ 0^t \circ \cdots \circ 0^t \circ c_k$.
return c

Notation. We introduce some additional notation we use throughout this section. We say that a set of integers $I \subset \mathbb{Z}$ of size n is an *interval* if $I = \{a, a+1, \ldots, a+(n-1)\}$ for some integer a. For two strings $x, y \in \{0,1\}^*$, we let $\text{LCS}(x,y) \in \{0,1\}^*$ denote the *longest common sub-sequence* between x and y. We also let $\mathbf{LCS}(x,y)$ denote a matching between x and y given by $\text{LCS}(x,y)$. Formally, $\mathbf{LCS}(x,y)$ denotes a sequence of tuples $(i_1, j_1), \ldots, (i_k, j_k)$, where $k = |\text{LCS}(x,y)|$ satisfying $i_1 < i_2 < \ldots < i_k \le |x|$, $j_1 < j_2 < \ldots j_k \le |y|$, and $x_{i_\ell} = y_{j_\ell}$ for all $\ell \in [k]$. Note that $\text{LCS}(x,y)$ need not be unique, but we can always fix one. It is well-known that $\mathsf{ED}(x,y) = |x| + |y| - 2 \cdot |\text{LCS}(x,y)|$.

6.1 Encoding Algorithm

The main ingredients of our encoding algorithm consist of an outer code $C_{\text{out}} \colon \{0,1\}^n \to \{0,1\}^k$ and an inner code C_{in} which we shall view as a mapping $C_{\text{in}} \colon [k] \times \{0,1\} \to \{0,1\}^t$. The encoding of $x \in \{0,1\}^n$ is given by

$$C(x) \coloneqq C_{\text{in}}(1, y_1) \circ 0^t \circ C_{\text{in}}(2, y_2) \circ 0^t \circ \cdots \circ 0^t \circ C_{\text{in}}(k, y_k) \ ,$$

where each $y_j \in \{0,1\}$ is obtained by writing $C_{\text{out}}(x) = y_1 \circ y_2 \circ \cdots \circ y_k$. Assuming the rate of C_{in} is a constant $r_{\text{in}} = \lceil 1 + \log_2 k \rceil / t > 0$, the length of this encoding $C(x)$ is $m \coloneqq (2k-1)t = \Theta(k \log(k))$. We note that the concatenation introduces buffers of 0s (i.e., the string 0^t), which slightly deviates from the standard code concatenation for Hamming errors. We present our formal encoding algorithm in Algorithm 1. Next we instantiate the concatenation framework by picking specific outer and inner codes.

The outer code. We take the outer code $C_{\text{out}} \colon \{0,1\}^n \to \{0,1\}^k$ to be a non-adaptive weak $(q_{\text{out}}, \delta_{\text{out}}, 1/2 + \varepsilon_{\text{out}})$-relaxed Hamming LDC given by Lemma 1 with $\gamma_{\text{out}} < \gamma$ and $\varepsilon_{\text{out}} = 2\varepsilon$, where γ and ε are given in Theorem 3. In particular, we have that $k = O(n^{1+\gamma_{\text{out}}})$ and $k \log(k) = O(n^{1+\gamma})$.

The inner code. We take the inner code $C_{\text{in}} \colon [k] \times \{0,1\} \to \{0,1\}^t$ to be an insertion-deletion code (i.e., it is a non-local code) due to Schulman and Zuckerman [SZ99], given by Lemma 2. In particular, C_{in} has the following properties:

1. C_{in} has constant rate $r_{\text{in}} = 1/\beta > 0$, where β is given in Lemma 2.
2. C_{in} has constant minimum (normalized) edit distance $\delta_{\text{in}} \in (0, 1/2)$.
3. For any interval $I \subseteq [t]$ with $|I| \ge 2$ and any $(j,y) \in [k] \times \{0,1\}$, it holds that $\mathsf{wt}(C_{\text{in}}(j,y)_I) \ge \lfloor |I|/2 \rfloor$, where $\mathsf{wt}(\cdot)$ denotes the Hamming weight.

With our choices of outer and inner codes, we prove the following key lemma about the resulting concatenation code C (i.e., Algorithm 1).

Lemma 9. *For any interval $I \subseteq [m]$ of length at most $(2 - \delta_{in})t$, and any index $j \in [k]$, we have* $\mathsf{ED}\left(C[I], C_{in}(j, 1 - y_j)\right) \ge \delta_{in} t/2$.

Remark 2. Note that for any interval I, we can lower bound the edit distance between $C[I]$ and $C_{\text{in}}(j, 1-y_j)$ by $\big||C_I| - |C_{\text{in}}(j, 1-y_j)|\big| = \big||I| - t\big|$. For any I of length greater than $(2-\delta_{\text{in}})t$, this implies a lower bound of $(1-\delta_{\text{in}})t \ge \delta_{\text{in}} t/2$ for all $\delta_{\text{in}} \le 1/2$.

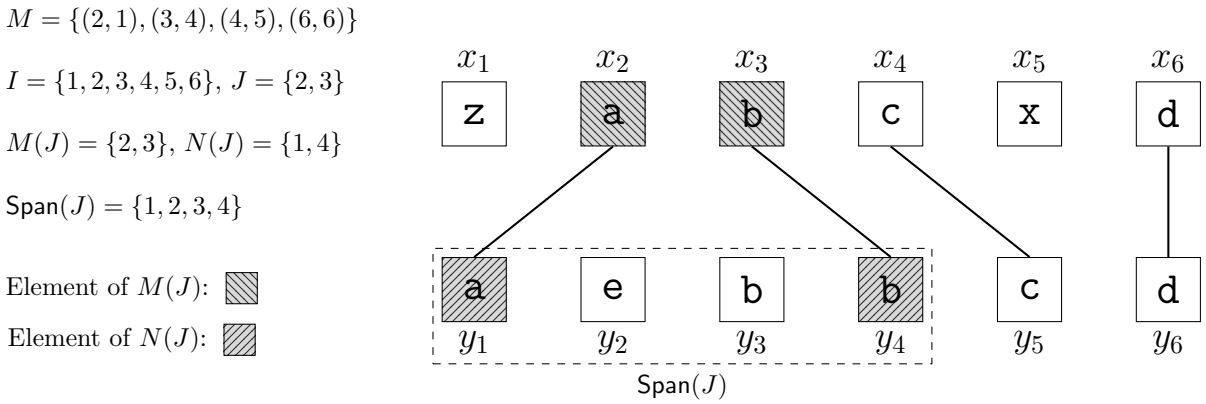


Figure 1: Pictorial representation of $M(J)$, $N(J)$, and $\mathsf{Span}(J)$. Here, $x = x_1 \circ \cdots x_6$ and $y = y_1 \circ \cdots y_6$, $\mathrm{LCS}(x,y) = \texttt{abcd}$ and $M = \mathbf{LCS}(x,y) = \{(2,1),(3,4),(4,5),(6,6)\}$. We take $I = \{1,\cdots,6\}$ and $J = \{2,3\}$ as an example. In the figure, the edges between nodes represent the matching M.

Proof. Consider an arbitrary LCS matching $M \coloneqq \mathbf{LCS}(C[I], C_{\text{in}}\left(j, 1-y_j\right)) \subseteq I \times [t]$ between $C[I]$ and $C_{\text{in}}\left(j, 1-y_j\right)$. We prove that $|M| \le (1-\delta_{\text{in}}/2)t$, from which the lemma follows.

We introduce some notation first. Given our LCS matching M and a interval $J \subseteq I$, we let $M(J) \coloneqq \{i \in J \colon \exists j \in [t], (i,j) \in M\}$ denote the set of indices in J that exist in the matching M. Let $N(J) \coloneqq \left\{j \in [t] \colon \exists i \in M(J), (i,j) \in M\right\} \subseteq [t]$ the set of indices in $[t]$ which are matched with $M(J)$. Finally, we denote by $\mathsf{Span}(J)$ the smallest interval that covers the set $N(J)$. We extend all of the definitions to unions of intervals as follows:[4]

$$M(J_1 \cup J_2) = M(J_1) \cup M(J_2),\ \ N(J_1 \cup J_2) = N(J_1) \cup N(J_2),\ \ \mathsf{Span}(J_1 \cup J_2) = \mathsf{Span}(J_1) \cup \mathsf{Span}(J_2).$$

See Figure 1 for a pictorial overview of $M(J)$, $N(J)$, and $\mathsf{Span}(J)$. Since M is a monotone matching, we have that $J_1 \cap J_2 = \varnothing$ implies $\mathsf{Span}(J_1) \cap \mathsf{Span}(J_2) = \varnothing$. We also have $|M(J)| = |N(J)| \le |\mathsf{Span}(J)|$.

Now we turn back to the proof. Since $|I| < 2t$, the interval I spans across at most 2 buffers and/or codewords. It is convenient to partition I into $I_b \cup I_c$, where I_b and I_c are unions of at most 2 intervals which correspond to buffers and codewords, respectively.

We first show that $\left|\mathsf{Span}(I_b)\right| \ge 2\left|M(I_b)\right|$. Intuitively, this is because the density of "1" is at least $1/2$ in any interval of an inner codeword, so every matched "0" in a buffer has to be accompanied with an insertion of "1". Formally, due to Property 3 of C_{in}, we have

$$\left|\mathsf{Span}(I_b)\right| \le 2\mathsf{wt}\Big(C_{\text{in}}(j, 1-y_j)[\mathsf{Span}(I_b)]\Big) \le 2\left|\mathsf{Span}(I_b) \setminus N(I_b)\right| = 2\left(\left|\mathsf{Span}(I_b)\right| - \left|M(I_b)\right|\right),$$

since any index $j \in N(I_b)$ is matched to an index in a buffer, which is necessarily a "0".

We finish the proof in two cases.

Case 1: I_c is the union of two intervals. In this case, we can deduce that I completely contains a buffer, i.e., $|I_b| = t$, and that $|I_c| \le |I| - |I_b| \le (1-\delta_{\text{in}})t$.

Therefore, we have

$$2|M| \le 2|M(I_c)| + 2|M(I_b)| \le |I_c| + \left|\mathsf{Span}(I_c)\right| + \left|\mathsf{Span}(I_b)\right| \le (1-\delta_{\text{in}})t + t = 2(1-\delta_{\text{in}}/2)t.$$

The last inequality is due to $\left|\mathsf{Span}(I_b)\right| + \left|\mathsf{Span}(I_c)\right| \le t$, since $\mathsf{Span}(I_b), \mathsf{Span}(I_c) \subseteq [t]$ are disjoint intervals.

Case 2: I_c is an interval. In this case, note that $M \cap (I_c \times [t])$ corresponds to a common subsequence between $C_{\text{in}}(j, 1-y_j)$ and some other codeword $C_{\text{in}}(j', y_{j'})$. The distance property of C_{in} implies $|M(I_c)| \le (1-\delta_{\text{in}})t$. Similarly we can upper bound $|M|$ by

$$2|M| = 2|M(I_c)| + 2|M(I_b)| \le (1-\delta_{\text{in}})t + \left|\mathsf{Span}(I_c)\right| + \left|\mathsf{Span}(I_b)\right| \le (1-\delta_{\text{in}})t + t = 2(1-\delta_{\text{in}}/2)t.$$

[4] $\mathsf{Span}(J_1 \cup J_2)$ may not be well-defined if $J_1 \cup J_2$ is itself an interval. However, in this paper we will only use this notation for disjoint and non-adjacent J_1 and J_2, in which case there is a unique way to partition $J_1 \cup J_2$ into disjoint intervals.

To conclude, we have $|M| \leq (1 - \delta_{\text{in}}/2)t$ in both cases. It follows that $\mathsf{ED}\left(C[I], C_{\text{in}}(j, 1 - y_j)\right) \geq \delta_{\text{in}} t/2$. □

6.2 The Decoding Algorithm

Our goal is to construct a relaxed decoder Dec that, given input an index $i \in [n]$ and oracle access to some binary string w which is δ-close to some codeword $C(x)$ in edit distance, outputs either the bit x_i or $\perp$ with probability at least $1/2 + \varepsilon$. We present our formal decoding algorithm in Algorithm 2.

In our construction, the decoding algorithm invokes the relaxed decoder for the outer code C_{out} while providing it with access to a simulated oracle $\widetilde{y} \in \{0,1\}^k$ using the oracle w which is the corrupted codeword. At a high level, the decoder operates as follows.

1. The decoder ensures that the oracle w has length m; else it outputs $\perp$.
2. The decoder invokes $C_{\text{out}}.\mathsf{Dec}(i)$.
3. For each query $j \in [k]$ received from $C_{\text{out}}.\mathsf{Dec}(i)$
   (a) The decoder computes codewords $C_{\text{in}}(j, 0)$ and $C_{\text{in}}(j, 1)$, and additionally computes the first index $i_0 \in [t]$ such that $C_{\text{in}}(j, 0)[i_0] \neq C_{\text{in}}(j, 1)[i_0]$.
   (b) The decoder samples $i_1, \ldots, i_d \in [t]$ uniformly and independently at random.
   (c) The decoder sets a bit $\widetilde{y}_j$ as follows. If $w[2(j-1)t + i_\ell] = C_{\text{in}}(j, 0)[i_\ell]$ for all $\ell \in \{0, 1, \ldots, d\}$, then set $\widetilde{y}_j = 0$; else if $w[2(j-1)t + i_\ell] = C_{\text{in}}(j, 1)[i_\ell]$ for all $\ell \in \{0, 1, \ldots, d\}$, then set $\widetilde{y}_j = 1$; else if neither case occurs, abort and output $\perp$.
   (d) Answer $C_{\text{out}}.\mathsf{Dec}(i)$ with bit $\widetilde{y}_j$ and await the next query.
4. Output symbol $\widetilde{x} = C_{\text{out}}.\mathsf{Dec}(i)$.

Remark 3. We choose to check that the received word has length m to simplify the analysis. However, this is not necessary by the following observations. First, if $|w| < m$, if the decoder every queries a symbol $j > |w|$, then we assume the oracle returns $\perp$, at which point our decoder can abort and output $\perp$. Second, if $|w| > m$, then our decoder will simply ignore any bits beyond w_m.

Analysis of the Decoder. Clearly, the query complexity of Algorithm 2 is $(d+1) \cdot q_{\text{out}} + 2$.[5] We take $\delta \coloneqq \delta_{\text{in}}\delta_{\text{out}}/128$. Fix a message $x \in \{0,1\}^n$ and oracle string w such that $\mathsf{ED}(C(x), w) \leq \delta \cdot 2m$. Moreover, let $y = C_{\text{out}}(x) \in \{0,1\}^k$. For $j \in [k]$, we denote by I_j the interval that correspond to $C_{\text{in}}(j, y_j)$ in $C(x)$. Formally,

$$I_j \coloneqq \left\{2(j-1)t + 1, \ldots, 2(j-1)t + t\right\}.$$

Let $\mathbf{LCS} \coloneqq \mathbf{LCS}(C(x), w)$. Note that $|\mathbf{LCS}| \geq (1 - \delta)m$ since $\mathsf{ED}(C(x), w) \leq \delta \cdot 2m$.

Definition 3. *We say $j \in [k]$ is* dangerous *if* $\mathsf{ED}\left(w[I_j], C_{in}(j, 1 - y_j)\right) \leq \delta_{in} t/4$.

We first show that if a block j is not dangerous, then $\widetilde{y}_j = 1 - y_j$ happens with small probability.

Proposition 2. *If j is not dangerous, then* $\Pr[\widetilde{y}_j = 1 - y_j] \leq (1 - \delta_{in}/8)^d$.

Proof. It suffices to show that for a uniformly random $i \in [t]$, we have

$$\Pr\left[w[2(j-1)t + i] = C_{\text{in}}(j, 1 - y_j)[i]\right] \leq 1 - \delta_{\text{in}}/8.$$

Since j is not dangerous, we have $\mathsf{ED}(w[I_j], C_{\text{in}}(j, 1 - y_j)) > \delta_{\text{in}} t/4$. Therefore

$$\begin{aligned}\Pr[w[2(j-1)t + i] \neq C_{\text{in}}(j, 1 - y_j)[i]] &= \frac{1}{t} \cdot \mathsf{HAM}\left(w[I_j], C_{\text{in}}(j, 1 - y_j)\right) \\ &\geq \frac{1}{2t} \cdot \mathsf{ED}(w[I_j], C_{\text{in}}(j, 1 - y_j)) > \frac{\delta_{\text{in}}}{8}.\end{aligned}$$

□

[5] If m is hard-coded in the decoder and the length m' of the received word is additionally given as input, then $q = (d+1) \cdot q_{\text{out}}$.

Algorithm 2: Decoding algorithm Dec.

Input : An index $i \in [n]$.
Oracle : Bitstring $w \in \{0,1\}^{m'}$ for some $m' \in \mathbb{N}$.
Output : A symbol $\widetilde{x} \in \{0,1,\perp\}$.
Hardcoded : Parameter $d \in \mathbb{N}$, outer decoder $C_{\text{out}}.\mathsf{Dec}\colon [k] \to \{0,1\}$, and inner encoder $C_{\text{in}}\colon [k] \times \{0,1\} \to \{0,1\}^t$.

if $w[m+1] \neq \perp$ *or* $w[m] = \perp$ **then return** $\widetilde{x} = \perp$ `/* Query Oracle. Verify length of` w. `*/`
foreach *oracle query* $j \in [k]$ *received from* $C_{out}.\mathsf{Dec}(i)$ **do** `/* Handle adaptive decoding. */`
  Compute $c_{j,0} = C_{\text{in}}(j,0)$ and $c_{j,1} = C_{\text{in}}(j,1)$.
  Compute first index $i_0 \in [t]$ such that $c_{j,0}[i_0] \neq c_{j,1}[i_0]$.
  Sample d values $i_1, i_2, \ldots, i_d \in [t]$ independently and uniformly at random.
  Compute `/* Query Oracle. Check consistency with computed inner codewords. */`

$$\widetilde{y}_j \coloneqq \begin{cases} 0 & \text{if } w[(j-1)\cdot 2t + i_\ell] = c_{j,0}[i_\ell]\ \forall \ell \in \{0,1,\ldots,d\} \\ 1 & \text{if } w[(j-1)\cdot 2t + i_\ell] = c_{j,1}[i_\ell]\ \forall \ell \in \{0,1,\ldots,d\} \\ \perp & \text{otherwise} \end{cases}$$

  if $\widetilde{y}_j = \perp$ **then return** $\widetilde{x} = \perp$.
  Answer query j of $C_{\text{out}}.\mathsf{Dec}(i)$ with value $\widetilde{y}_j$ and await the next query.
return $\widetilde{x} = C_{out}.\mathsf{Dec}(i) \in \{0,1,\perp\}$.

Now the key step is to upper bound the number of dangerous blocks.

Lemma 10. *The total number of dangerous blocks is at most* $\delta_{out}k/2$.

Proof. Let $\mathbf{LCS} \coloneqq \mathbf{LCS}(C(x), w)$ denote an arbitrary LCS matching between $C(x)$ and w. Let $U_C, U_w \subseteq [m]$ be the sets of unmatched bits in $C(x)$ and w, respectively; i.e., for every $i \in U_C$ (resp., $j \in U_w$), we have that $(i,\ell) \notin \mathbf{LCS}$ (resp, $(\ell, j) \notin \mathbf{LCS}$) for all $\ell \in [m]$. For each $j \in [k]$, define the following set of indices which are matched to I_j, i.e.,

$$N(I_j) \coloneqq \left\{ i \in [m] \colon \exists i' \in I_j, (i,i') \in \mathbf{LCS} \right\},$$

and let Span_j be the smallest interval covering $N(I_j)$. Since LCS is a monotone matching, the intervals $\mathsf{Span}_1, \ldots, \mathsf{Span}_k$ are disjoint. We can thus lower bound the edit distance between $C(x)$ and w by

$$\mathsf{ED}\left(C(x), w\right) = |U_C| + |U_w| \geq \sum_{j=1}^{k} \left| U_C \cap \mathsf{Span}_j \right| + \sum_{j=1}^{k} \left| U_w \cap I_j \right|.$$

Claim 5. *If* j *is dangerous, then either* $\left|U_C \cap \mathsf{Span}_j\right| \geq \delta_{in}t/8$, *or* $\left|U_w \cap I_j\right| \geq \delta_{in}t/16$.

Proof of Claim 5. Assume for the sake of contradiction that $\left|U_C \cap \mathsf{Span}_j\right| < \delta_{\text{in}}t/8$ and $|U_w \cap I_j| < \delta_{\text{in}}t/16$. We first note that

$$|\mathsf{Span}_j| = |U_C \cap \mathsf{Span}_j| + |N(j)| < \delta_{\text{in}}t/8 + t.$$

We also note that $\mathbf{LCS} \cap (\mathsf{Span}_j \times I_j)$ corresponds to a common subsequence between $C(x)[\mathsf{Span}_j]$ and $w[I_j]$, which has length at least

$$\left|\mathbf{LCS} \cap (\mathsf{Span}_j \times I_j)\right| = \left|I_j \setminus U_w\right| > t - \delta_{\text{in}}t/16.$$

In other words, we have

$$\begin{aligned} \mathsf{ED}\left(w[I_j], C(x)[\mathsf{Span}_j]\right) &< |I_j| + |\mathsf{Span}_j| - 2(t - \delta_{\text{in}}t/16) \\ &< t + (1 + \delta_{\text{in}}/8)t - 2t + \delta_{\text{in}}t/8 = \delta_{\text{in}}t/4. \end{aligned}$$

Since j is dangerous, we also have $\mathsf{ED}\left(w[I_j], C_{\text{in}}(j, 1-y_j)\right) \le \delta_{\text{in}} t/4$. The triangle inequality thus implies

$$\begin{aligned}\mathsf{ED}\left(C(x)[\mathsf{Span}_j], C_{\text{in}}(j, 1-y_j)\right) &\le \mathsf{ED}\left(w[I_j], C(x)[\mathsf{Span}_j]\right) + \mathsf{ED}\left(w[I_j], C_{\text{in}}(j, 1-y_j)\right) \\ &< \delta_{\text{in}} t/4 + \delta_{\text{in}} t/4 = \delta_{\text{in}} t/2.\end{aligned}$$

However, this contradicts Lemma 9. □

Denote by D the set of dangerous blocks. By Claim 5 we have

$$\delta \cdot 2m \ge \mathsf{ED}\left(w, C(x)\right) \ge \sum_{j \in D} \left(\left| U_C \cap \mathsf{Span}_j \right| + \left| U_w \cap I_j \right| \right) \ge |D| \cdot \frac{\delta_{\text{in}} t}{16}.$$

Plugging in $\delta = \delta_{\text{in}}\delta_{\text{out}}/128$ and $m = (2k-1)t \le 2kt$, we obtain $|D| \le \delta_{\text{out}} k/2$. □

Now we are ready to prove Theorem 3. We recall the theorem below.

Theorem 3. *For any $\gamma > 0$ and $\varepsilon \in (0, 1/2)$, there exist constants $\delta \in (0, 1/2)$ and $q = q(\delta, \varepsilon, \gamma)$, and non-adaptive weak $(q, \delta, 1/2 + \varepsilon)$-Insdel RLDCs $C \colon \{0,1\}^n \to \{0,1\}^m$ with $m = O(n^{1+\gamma})$.*

Proof. Consider the concatenation code C (with buffers) of Algorithm 1 and a relaxed decoder Dec defined in Algorithm 2. We fix an arbitrary message $x \in \{0,1\}^n$ and a string w such that $\mathsf{ED}\left(C(x), w\right) \le \delta \cdot 2m$. Here $\delta \coloneqq \delta_{\text{in}}\delta_{\text{out}}/128$. We also denote $y \coloneqq C_{\text{out}}(x) \in \{0,1\}^k$. For the remainder of the proof, unless otherwise stated, all lines referenced are from our decoder description in Algorithm 2.

Perfect completeness follows directly via the index i_0 computed in Algorithm 2, the computation of $\widetilde{y}_j$ in Algorithm 2, and the perfect completeness of the outer code C_{out}. We now focus on proving relaxed decoding.

Let $D \subseteq [k]$ be a subset containing the indices of all dangerous blocks in w. We have that $|D| \le \delta_{\text{out}} k/2$ by Lemma 10. Recall that in Algorithm 2, Dec invokes the relaxed decoder Dec_{out} of C_{out} while providing it with oracle access to some string $\widetilde{y} \in \{0,1\}^k$. Denote $\mathbf{Y} = (Y_1, Y_2, \ldots, Y_k)$ where each $Y_j \in \{0, 1, \perp\}$ is the result that would have been returned by the decoding algorithm on query j in Algorithm 2 of Algorithm 2 (even if j might not be queried). Since the decoder uses independent samples in Algorithm 2 of Algorithm 2 for each query j, Y_j's are independent random variables. Due to Proposition 2, for every $j \in [k] \setminus D$ it holds that

$$\Pr\left[Y_j = 1 - y_j\right] \le (1 - \delta_{\text{in}}/8)^d < e^{-d\delta_{\text{in}}/8} = \delta_{\text{out}}/4,$$

where in the last equality follows from choosing $d = 8\ln(4/\delta_{\text{out}})/\delta_{\text{in}}$.

Denote $d(\mathbf{Y}, y) = \sum_{j \in [k]} \mathbf{1}\left\{Y_j = 1 - y_j\right\}$. Since Y_j's are independent, an application of the Chernoff bound shows that

$$\Pr\left[d(\mathbf{Y}, y) > \delta_{\text{out}} k\right] \le \Pr\left[\sum_{j \in [k] \setminus D} \mathbf{1}\left\{Y_j = 1 - y_j\right\} \ge \frac{\delta_{\text{out}} k}{2}\right] \le \exp\left(-\delta_{\text{out}}^2 (k - |D|)/8\right) \le \varepsilon$$

for large enough n (and thus k). Given $\mathbf{Y} \in \{0, 1, \perp\}^k$, define a string $c(\mathbf{Y}) \in \{0,1\}^k$ as follows:

$$c(\mathbf{Y})_j \coloneqq \begin{cases} y_j & \text{if } Y_j \in \{y_j, \perp\} \\ 1 - y_j & \text{if } Y_j = 1 - y_j \end{cases}.$$

According to Algorithm 2 of Algorithm 2, Dec aborts with output $\perp$ as long as $Y_j = \perp$ for any query j. On the other hand, if $Y_j \neq \perp$ for all queries j, then Dec should output the result returned by Dec_{out} as if it had oracle access to $c(\mathbf{Y})$. In any case, we have

$$\Pr\left[\mathsf{Dec}(i, w) \in \{x_i, \perp\} \mid \mathbf{Y}\right] \ge \Pr\left[\mathsf{Dec}_{out}(i, c(\mathbf{Y})) \in \{x_i, \perp\}\right].$$

To conclude, we have

$$\begin{aligned}\Pr\left[\mathsf{Dec}(i,w)\in\{x_i,\perp\}\right] &= \mathbb{E}_{\mathbf{Y}}\left[\Pr\left[\mathsf{Dec}(i,w)\in\{x_i,\perp\}\mid\mathbf{Y}\right]\right]\\ &\geq \mathbb{E}_{\mathbf{Y}:\ d(\mathbf{Y},y)\leq\delta_{\text{out}}k}\left[\Pr\left[\mathsf{Dec}(i,w)\in\{x_i,\perp\}\mid\mathbf{Y}\right]\right]\cdot\Pr\left[d(\mathbf{Y},y)\leq\delta_{\text{out}}k\right]\\ &\geq \mathbb{E}_{\mathbf{Y}:\ d(\mathbf{Y},y)\leq\delta_{\text{out}}k}\left[\Pr\left[\mathsf{Dec}_{out}(i,c(\mathbf{Y}))\in\{x_i,\perp\}\right]\right]-\varepsilon.\end{aligned}$$

By definition of $d(\mathbf{Y},y)$, it holds that $\mathsf{HAM}(c(\mathbf{Y}),y)\leq\delta_{\text{out}}k$ whenever $d(\mathbf{Y},y)\leq\delta_{\text{out}}k$. Since C_{out} is a $(q_{\text{out}},\delta_{\text{out}},\varepsilon_{\text{out}})$-relaxed LDC, it holds that

$$\Pr\left[\mathsf{Dec}_{out}(i,c(\mathbf{Y}))\in\{x_i,\perp\}\right]\geq\frac{1}{2}+\varepsilon_{\text{out}}.$$

By our choice of $\varepsilon_{\text{out}}=2\varepsilon$, we have that

$$\Pr\left[\mathsf{Dec}(i,w)\in\{x_i,\perp\}\right]\geq\frac{1}{2}+2\varepsilon-\varepsilon=\frac{1}{2}+\varepsilon.$$

The query complexity is $q(\delta,\varepsilon,\gamma):=(d+1)\cdot q_{\text{out}}+2=\Theta\left(q_{\text{out}}\cdot\log(1/\delta_{\text{out}})/\delta_{\text{in}}\right)=\Theta(1)$. □

6.3 Strong Insdel Relaxed Locally Correctable Codes

In this section, we show how to tweak the above construction to obtain a strong Insdel relaxed Locally Correctable Code (RLCC) with constant locality when the outer Hamming RLDC is replaced with a Hamming RLCC. We first give a formal definition of relaxed Locally Correctable Codes.

Definition 4 (Relaxed Locally Correctable Code)**.** *A (q,δ,α,ρ)-Relaxed Locally Correctable Code (RLCC) $C\colon\Sigma^n\to\Sigma^m$ is a code for which there exists a randomized decoder that makes at most q queries to the received word y, and satisfies the following properties:*

1. *(Perfect completeness) For every $i\in[m]$, if $y=C(x)$ for some message x then the decoder, on input i, outputs y_i with probability* 1*.*
2. *(Relaxed decoding) For every $i\in[m]$, if y is such that* $\text{dist}(y,C(x))\leq\delta$ *for some unique $C(x)$, then the decoder, on input i, outputs c_i or $\perp$ with probability $\geq\alpha$.*
3. *(Success rate) For every y such that* $\text{dist}(y,C(x))\leq\delta$ *for some unique $C(x)$, there is a set I of size $\geq\rho m$ such that for every $i\in I$ the decoder, on input i, correctly outputs c_i with probability $\geq\alpha$.*

We will denote an RLCC that satisfies all 3 conditions by the notion of Strong RLCC, and one that satisfies the first 2 conditions by the notion of Weak RLCC. Furthermore, if the q queries are made in advance, before seeing entries of the codeword, then the decoder is said to be non-adaptive*; otherwise, it is called* adaptive*.*

As with Definition 1, the probabilities in the above definition are taken over the randomness of the decoding algorithm, and dist is a normalized metric. When dist is the normalized Hamming distance, then we say that the code is a Hamming RLCC; similarly, when dist is the normalized edit distance, we say that the code is a Insdel RLCC. To obtain our strong Insdel RLCC, we replace the weak Hamming RLDC of Algorithm 1 with the weak Hamming RLCC of Lemma 3 due to [AS21].

We now prove the following corollary.

Corollary 1. *For any $\gamma>0$ and $\varepsilon\in(0,1/2)$, there exist constants $\delta\in(0,1/2)$ and $q=q(\delta,\varepsilon,\gamma)$, and non-adaptive strong $(q,\delta,1/2+\varepsilon,1/2)$-Insdel RLCCs $C\colon\{0,1\}^n\to\{0,1\}^m$ with $m=O(n^{1+\gamma})$.*

Proof. The construction is based on a slight modification of code presented in Section 6.

For the encoding process, the outer code $C_{\text{out}}\colon\{0,1\}^n\to\{0,1\}^k$ is replaced by a non-adaptive weak $(q_{\text{out}},\delta_{\text{out}},1/2+\varepsilon_{\text{out}})$-relaxed Hamming LCC. The existence of such a code is guaranteed by Lemma 3. We pick $\varepsilon_{\text{out}}=2\varepsilon$ and $q_{\text{out}}(\delta,\varepsilon,\gamma)$ to be a sufficiently large constant such that $k=n^{1+O(1/q)}$ and $k\log k=O(n^{1+\gamma})$. The encoding algorithm is the same as Algorithm 1.

To ensure the total length of zero buffers is exactly half of the codeword length, we append another t 0's at the end of the codeword output by Algorithm 1. By the analysis in Section 6.1, we know that for any message $x\in\{0,1\}^n$, the codeword $C(x)$ has length $\Theta(k\log k)=O(n^{1+\gamma})$.

The decoding process is similar to Algorithm 2. Given an input index i, we do the following:

1. If the index i belongs to a zero buffer, the decoder aborts and outputs 0.
2. The decoder checks if the oracle w has length m. If not, it aborts and outputs $\perp$.
3. Let $\tilde{i}$ be the index of the non-zero block that contains i-th symbol. The decoder invokes $C_{\text{out}}.\mathsf{Dec}(\tilde{i})$.
4. For each query $j \in [k]$ received from $C_{\text{out}}.\mathsf{Dec}(\tilde{i})$
   (a) The decoder computes codewords $C_{\text{in}}(j,0)$ and $C_{\text{in}}(j,1)$, and additionally computes the first index $i_0 \in [t]$ such that $C_{\text{in}}(j,0)[i_0] \neq C_{\text{in}}(j,1)[i_0]$.
   (b) The decoder samples $i_1, \ldots, i_d \in [t]$ uniformly and independently at random.
   (c) The decoder sets a bit $\widetilde{y}_j$ as follows. If $w[2(j-1)t + i_\ell] = C_{\text{in}}(j,0)[i_\ell]$ for all $\ell \in \{0,1,\ldots,d\}$, then set $\widetilde{y}_j = 0$; else if $w[(j-1)(2t+1) + i_\ell] = C_{\text{in}}(j,1)[i_\ell]$ for all $\ell \in \{0,1,\ldots,d\}$, then set $\widetilde{y}_j = 1$; else if neither case occurs, abort and output $\perp$.
   (d) Answer $C_{\text{out}}.\mathsf{Dec}(i)$ with bit $\widetilde{y}_j$ and await the next query.
5. Let $\widetilde{y}_{\tilde{i}} = C_{\text{out}}.\mathsf{Dec}(\tilde{i})$. Compute the $\tilde{i}$-th non-zero block $C_{\text{in}}(\tilde{i}, \widetilde{y}_{\tilde{i}})$. Output the bit in this block that is in the i-th position of the whole codeword.

For the perfect completeness, we note our decoding algorithm will always output $C(x)_i$ if there are no corruption, which is guaranteed by the perfect completeness of the weak Hamming RLCC.

If the input index i is in one of the zero buffers, the decoder always outputs 0 correctly. The success rate for those indices is 1. The relaxed decoding property holds for those indices.

For i not in a zero buffer, let $\tilde{i}$ be the index of the non-zero block that contains the i-th symbol. By the relaxed decoding property of the weak Hamming RLCC, given access to $\mathbf{Y}$ (defined in the proof of Theorem 3), the relaxed decoder Dec_{out} of C_{out} will output $C_{\text{out}}(x)_{\tilde{i}}$ or $\perp$ with probability at least $1/2 + \varepsilon_{\text{out}}$. If Dec_{out} outputs $C_{\text{out}}(x)_{\tilde{i}}$ correctly, our decoder can also output $C(x)_i$ correctly. From the analysis in the proof of Theorem 3, we know the decoder will output $C(x)_i$ or $\perp$ with success probability at least $1/2 + \varepsilon$. Thus, the relaxed decoding property holds.

Finally, we can let $I_y \subseteq [m]$ be the set of all indices that is in a zero buffer. Since the total length of zero buffers is exactly half of the codeword length, $\left|I_y\right| = m/2$. We always output 0 correctly for $i \in I_y$. Thus, our code achieves a success rate $\rho \geq 1/2$. □

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